

Jindo: Practical Lattice-Based Polynomial Commitment for Zero-Knowledge Arguments

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Abstract. We present *Jindo*, a new lattice-based polynomial commitment scheme (PCS) readily available for compiling polynomial interactive oracle proofs (PIOP) into zero-knowledge arguments of knowledge (ZKAoK). *Jindo* improves upon the previous lattice-based PCSs *CELPC* (CRYPTO’ 24) and *Greyhound* (CRYPTO’ 24) by seamlessly integrating their strengths. Specifically, we incorporate the coefficient encoding method of *CELPC* with a new evaluation protocol inspired by *Greyhound*, inheriting only the advantages of both schemes. As a result, *Jindo* enjoys native support for a large base field, a large challenge set, the evaluation hiding property, and efficient batched evaluation.

Our implementation shows that *Jindo* improves *CELPC* with 1.3x faster proof generation, 3.7x faster verification, and 4.8x smaller proof size when evaluating polynomials of degree 2^{19} over a 255-bit prime field. Furthermore, *Jindo* outperforms *CELPC* with 3.5x faster proof generation, 9.7x faster verification, and 12.3x smaller proof size when compiling PIOP from *Buckler* (CCS’ 25) to prove the validity of an RLWE sample, demonstrating its practical efficacy in ZKAoK construction from PIOP.

Keywords: polynomial commitment; zero-knowledge argument; lattice

1 Introduction

A zero-knowledge argument of knowledge (ZKAoK) is a cryptographic protocol that allows a prover to convince a verifier that the prover knows a secret that makes a given statement true, without revealing any information beyond the statement’s validity. It has been a core building block for various privacy-preserving protocols, where the selective disclosure of a client’s input to a server is required for authentication while preserving the client’s privacy. Examples include e-voting [ABGS23], anonymous credentials [BLNS23], secure aggregation [BGL⁺23], and privacy-preserving signatures [AGJ⁺24]. In these protocols, the client’s computational overhead is primarily dominated by the overhead of proof generation. Therefore, improving the efficiency of ZKAoK is crucial for practical deployments of many privacy-preserving protocols.

In recent years, the efficiency of ZKAoKs has improved rapidly, driven by the polynomial interactive oracle proof (PIOP) framework [BFS20, CHM⁺20],

which enables a systematic and unified approach for constructing ZKAoKs. In the PIOP framework, a proof system is first designed as an interactive oracle proof [BSCS16], where in each round the prover sends oracles representing polynomials and the verifier responds with a random challenge or evaluation queries to the polynomial oracles. Such an IOP is then compiled into a ZKAoK using a polynomial commitment scheme (PCS), a variant of commitment scheme that enables efficient evaluation of committed polynomials over finite fields. For a proof system constructed via PIOP compilation, the efficiency of the PCS used in the compilation determines the overall performance, including proof generation, verification, and proof size. Therefore, improving the efficiency of PCSs used in PIOP compilations has been a major research focus for enhancing the overall performance of PIOP-based ZKAoKs.

For constructing efficient PCS, there have been various constructions based on different cryptographic assumptions. PCSs were initially designed using pairings [KZG10, Lee21] or discrete logarithms [BCC⁺16, BBB⁺18, WTS⁺18] based assumptions, leveraging the algebraic structure of elliptic curves. However, these constructions suffer from slow proof generation due to expensive elliptic curve operations and lack post-quantum security. As an alternative, hash-based constructions [GLS⁺23, ZXZS20, GLH⁺25] have emerged, offering faster proving performance via symmetric primitives alongside post-quantum security. Despite these advances, proof generation speed remains a significant bottleneck for the practical deployment of ZKAoKs in many scenarios. To address this, a new line of research focuses on constructing PCSs from lattice cryptography, leveraging the efficiency of lattice operations which rely on fast linear algebra over small integers.

To date, the most efficient lattice-based constructions are CELPC [HSS24] and Greyhound [NS24], which provide the fastest proving performance compared to hash or elliptic curve-based constructions. However, deploying these schemes for PIOP compilation remains challenging due to several practical limitations. CELPC prioritizes efficient support for large prime fields and the evaluation hiding property, enabling seamless PIOP compilation for ZKAoKs. However, it suffers from larger proof size due to a small challenge space, requiring multiple repetitions to achieve negligible soundness error. In contrast, Greyhound focuses on compact proof size and fast proving performance, achieving order-of-magnitude improvements over hash-based PCSs. However, it is highly optimized for small prime fields, and the concrete proof generation overhead of achieving evaluation hiding is not quantified, limiting its general applicability in PIOP compilation for ZKAoKs. Lastly, both PCSs support batched evaluation only in a limited sense, which is crucial for concrete efficiency in PIOP compilation. Thus, further improvements are required for lattice-based PCSs to enable efficient construction of ZKAoKs from PIOP compilations.

Scheme	Large Base Field ($\geq 2^{128}$)	Large Challenge Set ($\geq 2^{128}$)	Evaluation Hiding	Batched Evaluation
CELPC [HSS24]	O	X	O	$\triangle$
Greyhound [NS24]	X	O	$\triangle$	$\triangle$
Jindo (Ours)	O	O	O	O

Table 1: Comparison of lattice-based polynomial commitment schemes. “O” denotes full support, “ $\triangle$ ” denotes limited support, and “X” denotes inefficient or no support.

1.1 Our Contribution

In this work, we propose a new lattice-based polynomial commitment scheme, Jindo¹, resolving the limitations remaining in the previous constructions CELPC [HSS24] and Greyhound [NS24], which are summarized in Table 1. Interestingly, the strengths and limitations of these two schemes are complementary to each other, thus integrating both PCSs would yield an ideal lattice-based PCS construction. To do so, we observe that the overall structure of the polynomial evaluation protocols in CELPC and Greyhound is similar, as both are based on the Ajtai commitment scheme [Ajt96] and employ a sigma protocol for polynomial evaluation whose communication and verification costs follow square-root complexity.² However, differences in the detailed techniques used in each protocol lead to distinct characteristics in the resulting PCSs. Thus, our goal is to construct an efficient lattice-based PCS, eliminating previous limitations by seamlessly integrating the evaluation protocols from both schemes. Below, we briefly summarize our contributions in constructing the Jindo PCS.

Enhanced Evaluation Protocol. We observe that all shortcomings in the previous constructions stem from structural limitations in their evaluation protocols. Suppose we commit polynomials in $\mathbb{Z}_p[X]$ using the Ajtai commitment defined over $R_q = \mathbb{Z}_q[X]/(X^d + 1)$. For CELPC, it lacks a large challenge set because the knowledge soundness of its evaluation protocol relies on the extraction mechanism from [BCK⁺14], which limits the challenge set size to d . For Greyhound, its evaluation protocol supports a large challenge set but requires $p = q$, as its extraction method is based on [BS23], which restricts flexibility in base field choice. This is problematic because large p is needed for sufficient soundness in PIOP compilation, while small q is preferred for efficient PCS. Moreover, Greyhound requires additional overheads to achieve evaluation hiding, while CELPC natively supports it. In our PCS, we resolve all these issues simultaneously by redesigning the evaluation protocol. Our protocol supports evaluation hiding by default, following the approach of CELPC, and enables a challenge set of size p without requiring $p = q$, by developing a new extraction mechanism inspired by Greyhound. As a result, our protocol eliminates the inefficiency from repetition

¹ Named after the Jindo dog breed, following the convention of Greyhound and Labrador.

² Here, we exclude the Labrador layer of GreyHound, as it can be applied to both PCS.

by using a large challenge set, while maintaining the evaluation hiding property and flexibility in base field choice, as detailed in Table 1.

Batched Evaluation via Homomorphism. PIOP compilations for ZKAoKs often require evaluating multiple polynomials simultaneously in a batched manner. Thus, beyond improving the core evaluation protocol, we construct an efficient batched evaluation protocol for Jindo by leveraging its homomorphic properties. In discrete logarithm or pairing-based PCSs, taking a random linear combination of commitments is a well-established optimization technique for reducing both proof size and verification costs in batched evaluation. Although homomorphic properties are inherent in lattice-based PCSs, they are relatively underutilized due to structural limitations in previous constructions. We observe that our new PCS Jindo efficiently supports homomorphism for batched evaluation, and adapt the random linear combination technique to the lattice setting within Jindo, requiring operations only at the commitment level while keeping the evaluation and verification algorithms unchanged. This results in further reductions in proof size and verification overhead when evaluating multiple polynomials simultaneously.

Implementation. To demonstrate concrete efficiency, we provide an implementation of our PCS. Compared to CELPC, we achieve a $1.3\times$ reduction in proof generation time, a $4.8\times$ smaller proof size, and a $3.7\times$ faster verification cost, primarily due to the larger challenge set, which eliminates the need for protocol repetition. We also report the performance gains from batched evaluation. When evaluating 32 polynomials, we achieve a $13\times$ reduction in proof size and a $12.5\times$ reduction in verification cost, demonstrating the impact of homomorphism. Finally, we benchmark our PCS within PIOP compilation scenarios to showcase its impact on ZKAoK construction. We measure the performance of proving the validity of RLWE samples, using the PIOP-based ZKAoK constructions from [HLSS25]. The results show that Jindo achieves $3.6\times$ faster proof generation, $12.3\times$ smaller proof sizes, and $9.7\times$ faster verification compared to compilations using CELPC. These improvements stem from comprehensive optimizations in the enhanced polynomial evaluation protocol and the effective utilization of homomorphism for batched evaluation, confirming the practicality of our PCS for PIOP-based ZKAoK constructions.

1.2 Related Work

Over the past decade, lattice-based proof systems have been extensively studied from various directions. The most relevant line of work to ours is the construction [HMSS25a, HMSS25b, HLSS25] of zero-knowledge arguments of knowledge (ZKAoKs) for lattice-based homomorphic encryption (HE) schemes via PIOP compilations using CELPC. These works achieve practical performance for generating the necessary proofs in HE-based protocols by leveraging the generality of the PIOP framework, but proof size remains the primary bottleneck due to reliance on CELPC. Besides the PIOP framework, Labrador [BS23] constructs an efficient proof system capable of proving R1CS satisfiability directly from the

Ajtai commitment scheme. While **Labrador** offers compact, polylogarithmic proof sizes, it suffers from linear verification complexity. Furthermore, it is highly optimized for specific parameter sets, and the concrete overhead required to achieve zero-knowledge is not documented in the literature, limiting its general applicability in ZKAoK constructions.

Another research direction [ALS20, ENS20, LNS20, LNP22] builds proof systems from the BDLOP commitment scheme [BDL⁺18]. These works focus on constructing efficient ZKAoKs for lattice-based primitives, demonstrating their practicality through the **Lazer** library [LSS24]. However, since BDLOP commitments grow linearly with the input size, their efficiency is limited to applications such as lattice-based signatures or KEMs, where input dimensions are small. More recently, a growing line of research [ACL⁺22, FMN24, KLNO24] builds proof systems from non-standard lattice assumptions, focusing on improving verification complexity. However, the practical efficiency of these schemes in constructing ZKAoKs, including the overhead of achieving zero-knowledge and proof generation speed, has not yet been presented.

2 Technical Overview

In this section, we provide a technical overview of our new PCS, **Jindo**. A more detailed and formal analysis is presented in the subsequent sections.

2.1 Ajtai Commitment

Our starting point is the variant of the Ajtai commitment scheme introduced in CELPC, which allows committing to coefficient from a large prime field $\mathbb{Z}_p$. The basic structure is as follows, defined over a polynomial ring $R_q = R/qR$, where $R = \mathbb{Z}[X]/(X^d + 1)$.

$$\mathbf{A}\mathbf{m} + \mathbf{B}\mathbf{r} = \mathbf{t} \pmod{q}$$

In the above, $\mathbf{A}$ and $\mathbf{B}$ are uniformly random matrices over R_q , serving as the commitment key. $\mathbf{m}$ and $\mathbf{r}$ are vectors over R , corresponding to the input message and the randomness. Lastly, the commitment $\mathbf{t}$ is a vector defined over R_q , which has much smaller dimension compared to $\mathbf{m}$, thereby compressing the input message.

Hiding & Biding. The hiding property is derived from the Module Learning With Errors (MLWE) problem, assuming $\mathbf{B}\mathbf{r}$ is computationally indistinguishable from a uniformly random sample. To ensure the binding property, which is based on the Module Short Integer Solution (MSIS) problem, the vectors $\mathbf{m}$ and $\mathbf{r}$ must have small norms.

Encoding for Large Prime Field. Due to the norm constraints on $\mathbf{m}$, coefficients in $\mathbb{Z}_p$ cannot be directly committed when $p \gg q$. To resolve this, we utilize the following encoding method, introduced in CELPC, which is derived

from the isomorphism $\mathbb{Z}_p \cong R/(X - b)R$ when $p = b^d + 1$.

$$\text{Ecd} : a = \sum_{j=0}^{d-1} a_j b^j \in \mathbb{Z}_p \mapsto m(X) = \sum_{j=0}^{d-1} a_j X^j \in R$$

In the above, each a_j can be bounded by $b \ll q$. Thus, a vector $\mathbf{a} \in \mathbb{Z}_p^k$ can be encoded into a vector $\mathbf{m} \in R^k$, whose norm is much smaller than q , by applying the above encoding to each coefficient.

The other key advantage of the above map is its homomorphic property, which will be utilized for batched evaluation. Define the decoding map as follows.

$$\text{Dcd} : m(X) \in R \mapsto m(b) \in \mathbb{Z}_p$$

Then, for any $a, b, c \in \mathbb{Z}_p$, the following homomorphic property holds.

$$ab + c = \text{Dcd}(\text{Ecd}(a) \cdot \text{Ecd}(b) + \text{Ecd}(c))$$

2.2 Polynomial Evaluation Protocol

We aim to simultaneously address the previous limitations related to the base field size, challenge set, and evaluation hiding property by constructing an enhanced polynomial evaluation protocol.

Basic Structure. We follow the basic structure of the polynomial evaluation protocol in Greyhound, while adapting the encoding method from CELPC. To be precise, let $f(X) = \sum_{i=0}^{N-1} f_i X^i \in \mathbb{Z}_p^{<N}[X]$ be a polynomial of degree bound $N = mn$. Then, its evaluation $y = f(x)$ at a point $x \in \mathbb{Z}_p$ can be represented as follows.

$$y = [1 \ x^n \ \dots \ x^{(m-1)n}] \begin{bmatrix} f_0 & \dots & f_{n-1} \\ \vdots & & \vdots \\ f_{(m-1)n} & \dots & f_{N-1} \end{bmatrix} \begin{bmatrix} 1 \\ x \\ \vdots \\ x^{n-1} \end{bmatrix}$$

However, since the message space of the Ajtai commitment consists of short elements in R , we rewrite the above equation by leveraging the encoding method from CELPC, which can be described as follows:

$$y = \text{Dcd} \left(\hat{\mathbf{x}}^\top \left[\hat{\mathbf{f}}_0 \mid \dots \mid \hat{\mathbf{f}}_{n-1} \right] \tilde{\mathbf{x}} \right) \quad (1)$$

where $\hat{\mathbf{x}} \in R^n$, $\tilde{\mathbf{x}} \in R^m$, and $\hat{\mathbf{f}}_j \in R^m$ are defined as follows.

$$\begin{aligned} \hat{\mathbf{x}} &:= \left(1, \text{Ecd}(x^n), \dots, \text{Ecd}(x^{(m-1)n}) \right) \\ \tilde{\mathbf{x}} &:= \left(1, \text{Ecd}(x), \dots, \text{Ecd}(x^{n-1}) \right) \\ \hat{\mathbf{f}}_j &:= \left(\text{Ecd}(f_j), \text{Ecd}(f_{n+j}), \dots, \text{Ecd}(f_{(m-1)n+j}) \right) \end{aligned}$$

Next, we achieve square-root communication and verification cost for evaluating Eq. (1) using a three-round sigma protocol. In the first round, the prover

sends the commitments $\mathbf{t}_0, \dots, \mathbf{t}_{n-1} \in R^\mu$ and the partial evaluation results $\mathbf{z} \in R^n$, which are generated as follows:

$$\mathbf{t}_j := \mathbf{A}\hat{\mathbf{f}}_j + \mathbf{B}\mathbf{r}_j, \quad \mathbf{z} := \hat{\mathbf{x}}^\top \begin{bmatrix} \hat{\mathbf{f}}_0 \\ \vdots \\ \hat{\mathbf{f}}_{n-1} \end{bmatrix}$$

where $\mathbf{r}_j \in R^{\mu+\nu}$ are commitment randomness for the hiding property.

In the second round, the verifier replies with short challenges defined as follows.

$$\mathbf{c} = (c_0, \dots, c_{n-1}) \in R^n$$

In the final round, the prover responds with $\bar{\mathbf{f}} \in R^m$ and $\bar{\mathbf{r}} \in R^{\mu+\nu}$ defined as follows.

$$\bar{\mathbf{f}} := \sum_{j=0}^{n-1} c_j \hat{\mathbf{f}}_j, \quad \bar{\mathbf{r}} := \sum_{j=0}^{n-1} c_j \mathbf{r}_j,$$

Then, the verifier checks whether $\bar{\mathbf{r}}, \bar{\mathbf{f}}$ are short and the following equations hold.

$$\mathbf{A}\bar{\mathbf{f}} + \mathbf{B}\bar{\mathbf{r}} \stackrel{?}{=} \sum_{j=0}^{n-1} c_j \mathbf{t}_j, \quad \hat{\mathbf{x}}^\top \bar{\mathbf{f}} \stackrel{?}{=} \mathbf{z}^\top \mathbf{c}, \quad \text{Dcd}(\mathbf{z}^\top \hat{\mathbf{x}}) \stackrel{?}{=} y \quad (2)$$

Both the communication cost and the verification cost roughly follow $O(m+n)$, assuming the parameters μ and ν related to MSIS and MLWE are almost constants. Thus, the optimum of both complexities is $O(\sqrt{N})$, as $mn = N$.

Randomized Encoding. Unfortunately, the evaluation protocol described so far does not achieve the evaluation hiding property, because $\mathbf{z}$ and $\bar{\mathbf{f}}$ leak more information about the input polynomial than the evaluation result $y = f(x)$. Additionally, $\bar{\mathbf{r}}$ leaks some information about the commitment randomness, which compromises the hiding property. To resolve this, we integrate the randomized encoding technique from CELPC into our protocol. To prevent unnecessary information leakage, we need to add blind factors to $\bar{\mathbf{f}}, \bar{\mathbf{r}}$, and $\mathbf{z}$. However, for the Ajtai commitment, the masking procedure must preserve smallness to maintain the binding property, which requires nontrivial treatment.

The randomized encoding technique introduced in CELPC successfully resolves this issue. The core idea is to randomize the encoding map Ecd into REcd as follows,

$$\text{REcd}_{\sqrt{\Sigma}} : a \in \mathbb{Z}_p \mapsto m \stackrel{\$}{\leftarrow} \mathcal{D}_{\text{Ecd}(a)+\Lambda, \sqrt{\Sigma}}$$

where $\text{Ecd}(a) + \Lambda \subseteq R$ is the set of polynomials congruent to $\text{Ecd}(a)$ modulo $(X - b)$, forming a coset of some lattice Λ , and $\mathcal{D}_{\text{Ecd}(a)+\Lambda, \sqrt{\Sigma}}$ is the discrete Gaussian distribution defined over this coset with covariance matrix $\sqrt{\Sigma}$. Then, we can replace Ecd with REcd while maintaining evaluation correctness, as $\text{Dcd}(\text{REcd}_{\sqrt{\Sigma}}(a)) = a$ for all $a \in \mathbb{Z}_p$.

The main advantage of randomized encoding is that it greatly simplifies the masking procedure in R , analogous to that in $\mathbb{Z}_p$. For example, to mask

a value $a \in \mathbb{Z}_p$, we typically add a uniformly random sample $u \xleftarrow{\$} \mathbb{Z}_p$, since the distribution of $a + u$ is also uniformly random. Similarly, to mask a value $m \xleftarrow{\$} \text{REcd}_{\sqrt{\Sigma}}(a)$, we can add $m' \xleftarrow{\$} \text{REcd}_{\sqrt{\Sigma'}}(u)$. Then, by the convolution lemma [Pei10], the distribution of $m + m'$ is statistically close to $\mathcal{D}_{\text{Ecd}(a+u)+\Lambda, \sqrt{\Sigma+\Sigma'}}$, which is easily simulatable since $a + u$ follows a uniform distribution and Σ and Σ' are public values.

Evaluation Hiding. Based on this property, we can first design a masking procedure in $\mathbb{Z}_p$ and then lift it to R by applying randomized encoding to all elements in $\mathbb{Z}_p$, which results in achieving the evaluation hiding property. First, let F be the following augmented matrix for the witness polynomial $f(X) = f_0 + f_1X + \dots + f_{N-1}X^{N-1}$,

$$F = \begin{bmatrix} f_0 & f_1 - b_1 & \dots & f_{n-2} - b_{n-2} & f_{n-1} - b_{n-1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ f_{(m-1)n} & f_{(m-1)n+1} & \dots & f_{N-2} & f_{N-1} \\ b_1 & b_2 & \dots & b_{n-1} & 0 \end{bmatrix}$$

where $b_1, \dots, b_{n-1} \xleftarrow{\$} \mathbb{Z}_p$. Then, the following still holds for the augmented matrix F by modifying the left-hand side vector:

$$f(x) = [1 \ x^n \ \dots \ x^{(m-1)n} \ x] F \begin{bmatrix} 1 \\ x \\ \vdots \\ x^{n-1} \end{bmatrix}$$

Then, the evaluation of $\mathbf{z}$ corresponds to the following evaluation in $\mathbb{Z}_p$.

$$[y_0 \ \dots \ y_{n-1}] := [1 \ x^n \ \dots \ x^{(m-1)n} \ x] F$$

We note that the distribution of $(y_0, \dots, y_{n-1})$ follows the uniform distribution over $\mathbb{Z}_p^n$ whose inner product with $(1, x, \dots, x^{n-1})$ is y , due to the blind factors $b_1, \dots, b_{n-1}$. Thus, if we generate $\hat{\mathbf{f}}_j$ via randomized encoding of each column of F , and redefine $\hat{\mathbf{x}}$ to $(1, \text{Ecd}(x^n), \dots, \text{Ecd}(x^{(m-1)n}), \text{Ecd}(x))$, then $\mathbf{z}$ become simulatable using only the value of y .

Next, we redefine $\bar{\mathbf{f}}$ and $\bar{\mathbf{r}}$ as follows to include blinding factors $\hat{\mathbf{f}}_n$ and $\mathbf{r}_n$:

$$\bar{\mathbf{f}} := \sum_{j=0}^{n-1} c_j \hat{\mathbf{f}}_j + \hat{\mathbf{f}}_n, \quad \bar{\mathbf{r}} := \sum_{j=0}^{n-1} c_j \mathbf{r}_j + \mathbf{r}_n \quad (3)$$

where $\hat{\mathbf{f}}_n$ is a randomized encoding of $\mathbf{u} \xleftarrow{\$} \mathbb{Z}_p^{m+1}$ and $\mathbf{r}_n$ is sampled from a discrete Gaussian distribution. Then, $\bar{\mathbf{f}}$ becomes simulatable thanks to the convolution lemma, and $\bar{\mathbf{r}}$ does not reveal information on $\mathbf{r}_j$ when they are also sampled from a discrete Gaussian distribution, due to the Hint-MLWE problem [KLSS23].

New Extraction Method. To achieve a large challenge set, we develop a new witness extraction technique that extends the extraction technique used in Greyhound. In general, to extract exact opening knowledge for a commitment $\mathbf{t}$, the

challenge set, where each c_j is sampled, must consist of short elements in R whose pairwise differences are invertible in R_q , with the corresponding inverses also being short. However, finding such a large challenge set is notoriously difficult, as analyzed in [AL21]. In *Greyhound*, it is observed that weak opening knowledge for $\mathbf{t}$, i.e., the ability to open $\bar{c} \cdot \mathbf{t}$ for some short $\bar{c} \in R$, is sufficient for the binding property of the PCS. This observation relaxes the conditions on the challenge set, requiring only that each pairwise difference $\bar{c}$ be invertible in R_q , which allows for a larger challenge set depending on the algebraic structure of R_q . However, this comes at the cost of inflexibility in choosing the base field, since $p = q$ must hold, as coefficients are directly recovered in R_q by multiplying the inverse of $\bar{c}$.

In our evaluation protocol, we observe that the invertibility condition is refined to the invertibility of $\text{Dcd}(\bar{c})$ over $\mathbb{Z}_p$, thanks to the CELPC encoding method, eliminating the dependency between p and q . The detailed witness extraction procedure in *Jindo* is as follows. Similar to *Greyhound*, the extraction can be done using the coordinate-wise special soundness framework [FMN24]. For each $0 \leq j < n$, an extractor can obtain short $\bar{\mathbf{f}}_j$, $\bar{\mathbf{r}}_j$, and $\bar{c}_j$ satisfying the following by rewinding the second round.

$$\mathbf{A}\bar{\mathbf{f}}_j + \mathbf{B}\bar{\mathbf{r}}_j = \bar{c}_j \cdot \mathbf{t}_j, \quad \hat{\mathbf{x}}^\top \bar{\mathbf{f}}_j = \bar{c}_j \cdot z_j$$

Next, we assume $\text{Dcd}(\bar{c}_j) \neq 0$ so that $\text{Dcd}(\bar{c}_j)$ is invertible in $\mathbb{Z}_p$. This can be assured with overwhelming probability if p is sufficiently large and the c_j are sampled from $\text{Ecd}(\mathbb{Z}_p)$. Then, we can reconstruct a witness polynomial f satisfying $f(x) = y$ as follows, where $\hat{\mathbf{f}}_j = (\hat{f}_{0,j}, \dots, \hat{f}_{m-1,j})$.

$$f_{i,j} := \text{Dcd}(\bar{c}_j)^{-1} \cdot \text{Dcd}(\hat{f}_{i,j}), \quad f(X) := \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} f_{i,j} X^{ni+j}$$

As a result, an extractor can recover the input polynomial without relying on the algebraic structure of R_q , while achieving a sufficiently large challenge set to eliminate the need for repetition.

2.3 Batched Evaluation Protocol

To further improve the efficiency of our PCS in PIOP compilations, we provide a systematic analysis of the batched evaluation protocol via random linear combination in the lattice setting, extending the previous approaches [BDFG21, KST22, BCL⁺21] in discrete logarithm-based PCSs.

Relaxed Reduction of Knowledge. The batched evaluation protocol can be best understood in the reduction of knowledge (RoK) framework [KP23]. To apply the RoK framework to our PCS, we first generalize its definition to relaxed reduction of knowledge, allowing it to handle the peculiarities of lattice-based proof systems. In short, an RoK protocol is an interactive protocol that reduces a statement $(\mathbf{x}_1, \mathbf{w}_1) \in \mathbf{R}_1$ to a statement $(\mathbf{x}_2, \mathbf{w}_2) \in \mathbf{R}_2$, enabling a prover to prove $(\mathbf{x}_2, \mathbf{w}_2) \in \mathbf{R}_2$ instead of $(\mathbf{x}_1, \mathbf{w}_1) \in \mathbf{R}_1$ after the reduction. However, for

lattice-based proof systems, there is a gap between the relation R that an actual statement satisfies and the relation $R' \supseteq R$ that a verifier can verify, which inhibits the direct application of the RoK framework. Thus, we generalize the definition of RoK to accommodate this case, enabling a robust analysis of the batched evaluation protocol in the lattice setting.

Outer Commitment. In batched evaluation scenarios, the size of the evaluation proof remains the same as in a single evaluation due to the random linear combination technique, while the size of the commitments remains unchanged. Thus, it is crucial to reduce the size of the commitments to enhance the overall efficiency of the batched evaluation protocol. To resolve this, we adopt the outer commitment technique introduced in Greyhound, which was originally used for efficient proof composition with Labrador [BS23], but we repurpose it for efficient batched evaluations.

In short, an outer commitment $\mathbf{u}$ is a commitment to the commitments $\mathbf{t}_0, \dots, \mathbf{t}_{n-1}$, which we refer to as the inner commitments. More precisely, the outer commitment $\mathbf{u}$ is generated as follows using the Ajtai commitment,

$$\mathbf{u} := \mathbf{D}(\hat{\mathbf{t}}_0 \| \dots \| \hat{\mathbf{t}}_{n-1})$$

where $\mathbf{D}$ is a public matrix and each $\hat{\mathbf{t}}_j$ is a vector of short elements in R derived from $\mathbf{t}_j$. In Greyhound, $\hat{\mathbf{t}}_j$ is obtained from $\mathbf{t}_j$ via digit decomposition to reduce norm, at the cost of increased dimension. In contrast, we generate $\hat{\mathbf{t}}_j := \lfloor \mathbf{t}_j / q' \rfloor$ by rounding with respect to some q' , motivated by the commitment compression technique used in [DKL⁺18, LNS20, ENS20]. This effectively reduces the norm of each $\mathbf{t}_j$ while preserving dimension, thereby saving computation cost in generating the outer commitment. As a result, we significantly reduce the overall communication cost when batched evaluation is applied by replacing the inner commitments $\mathbf{t}_0, \dots, \mathbf{t}_{n-1}$ with the single outer commitment $\mathbf{u}$ in random linear combinations.

Random Linear Combinations. The batched evaluation protocol for our PCS is as follows. Suppose we evaluate ℓ polynomials $f_0, \dots, f_{\ell-1}$, associated with the outer commitments $\mathbf{u}_0, \dots, \mathbf{u}_{\ell-1}$. Our protocol aggregates them into a single outer commitment $\mathbf{u}$ using the homomorphic property:

$$\mathbf{u} := \sum_{k=0}^{\ell-1} \text{Ecd}(\alpha_k) \cdot \mathbf{u}_k$$

where each $\alpha_k \xleftarrow{\$} \mathbb{Z}_p$. Then, it suffices to generate a single evaluation proof for $\mathbf{u}$ with respect to the polynomial $f = \sum_{k=0}^{\ell-1} \alpha_k \cdot f_k$, instead of ℓ separate evaluation proofs for $\mathbf{u}_0, \dots, \mathbf{u}_{\ell-1}$, which effectively reduces the proof size and verification cost. For the security analysis, we prove that this protocol is a reduction of knowledge protocol that reduces ℓ polynomial evaluation relations into a single evaluation relation, based on the homomorphic property of our PCS.

3 Preliminaries

3.1 Notation

For a positive integer q , we use $\mathbb{Z} \cap (-q/2, q/2]$ as a representative set of $\mathbb{Z}_q$, and denote by $[a]_q$ the reduction of a modulo q . We regard all vectors as column vectors, and use the symbol $\parallel$ for the concatenation of two vectors. Let d be a power of two. We denote by $R = \mathbb{Z}[X]/(X^d + 1)$ the ring of integers of the $2d$ -th cyclotomic field, and by $R_q = \mathbb{Z}_q[X]/(X^d + 1)$ the residue ring of R modulo q . For a vector $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{Z}^n$, the L^p ($p \geq 1$) and L^∞ norms are defined as $\|\mathbf{v}\|_p := \sqrt[p]{\sum_{i=1}^n |v_i|^p}$ and $\|\mathbf{v}\|_\infty := \max_{1 \leq i \leq n} |v_i|$, respectively. For a polynomial f , $\|f\|_p$ is defined over its coefficient vector. For a matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, the matrix norm of $\mathbf{A}$ is defined by $\|\mathbf{A}\|_2 := \max_{\mathbf{x} \in \mathbb{R}^n \setminus \{0\}} \frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2}$. For a security parameter λ , we denote a negligible function with respect to λ as $\text{negl}(\lambda)$.

3.2 Discrete Gaussian Distribution

For a positive definite matrix $\Sigma \in \mathbb{R}^{n \times n}$, we define the elliptical Gaussian function $\rho_{\sqrt{\Sigma}} : \mathbb{R}^n \rightarrow (0, 1]$ as $\rho_{\sqrt{\Sigma}}(\mathbf{x}) := \exp(-\pi \cdot \mathbf{x}^\top \Sigma^{-1} \mathbf{x})$. Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and $\mathbf{c} \in \mathbb{R}^n$. The discrete Gaussian distribution $\mathcal{D}_{\mathbf{c}+\Lambda, \sqrt{\Sigma}}$ is defined as a distribution over the coset $\mathbf{c} + \Lambda$, whose probability mass function is $\mathcal{D}_{\mathbf{c}+\Lambda, \sqrt{\Sigma}}(\mathbf{x}) = \rho_{\sqrt{\Sigma}}(\mathbf{x}) / \rho_{\sqrt{\Sigma}}(\mathbf{c} + \Lambda)$ for $\mathbf{x} \in \mathbf{c} + \Lambda$ where $\rho_{\sqrt{\Sigma}}(\mathbf{c} + \Lambda) := \sum_{\mathbf{v} \in \mathbf{c} + \Lambda} \rho_{\sqrt{\Sigma}}(\mathbf{v}) < \infty$. When $\Sigma = \sigma^2 \cdot \mathbf{I}_n$ for $\Sigma > 0$ where $\mathbf{I}_n$ is the n -dimensional identity matrix, then we substitute $\sqrt{\Sigma}$ by σ in the subscript and refer to σ as the width parameter.

Definition 1 (Definition 3.1 [MR07]). For an n -dimensional lattice Λ and positive real $\varepsilon > 0$, the smoothing parameter $\eta_\varepsilon(\Lambda)$ is the smallest s such that $\rho_{1/s}(\Lambda^* \setminus \{0\}) \leq \varepsilon$.

Definition 2 (Definition 2.3 [Pei10]). Let Σ be a positive-definite matrix. We say that $\sqrt{\Sigma} \geq \eta_\varepsilon(\Lambda)$ if $\eta_\varepsilon(\sqrt{\Sigma}^{-1} \cdot \Lambda) \leq 1$.

Lemma 1 (Lemma 3.3 [MR07]). For any n -dimensional lattice Λ and $\varepsilon > 0$,

$$\eta_\varepsilon(\Lambda) \leq \sqrt{\frac{\ln(2n(1 + 1/\varepsilon))}{\pi}} \cdot \lambda_n(\Lambda)$$

where $\lambda_n(\Lambda)$ is the smallest real number $r > 0$ such that $\dim(\text{span}(\Lambda \cap \mathcal{B}(r))) = n$ and $\mathcal{B}(r)$ is the n -dimensional ball with radius r centered at the origin.

3.3 MSIS and MLWE

We review the hard problems in lattice cryptography that underlie our PCS.

Definition 3 (MSIS). Let μ, ℓ be positive integers, and $0 < \beta < q$. We say that a PPT adversary $\mathbf{A}$ has advantage ε in solving $\text{MSIS}_{q, \mu, \ell, \beta}$ if

$$\Pr \left[[\mathbf{A} \mid \mathbf{I}_\mu] \mathbf{x} = \mathbf{0} \pmod{q} \wedge \|\mathbf{x}\|_2 < \beta \mid \mathbf{A} \xleftarrow{\$} R_q^{\mu \times \ell}, \mathbf{x} \leftarrow \mathbf{A}(\mathbf{A}) \right] \geq \varepsilon.$$

Definition 4 (MLWE). Let ν, ℓ be positive integer, and χ be a distribution over R . We say that a PPT adversary $\mathbf{A}$ has advantages ε in solving $\text{MLWE}_{q,\nu,\ell,\chi}$ if

$$\left| \Pr \left[b = 1 \mid \mathbf{A} \xleftarrow{\$} R_q^{\ell \times \nu}, \mathbf{r} \xleftarrow{\$} \chi^{\nu+\ell}, b \leftarrow \mathbf{A}(\mathbf{A}, [\mathbf{A} | \mathbf{I}_\ell] \mathbf{r}) \right] - \Pr \left[b = 1 \mid \mathbf{A} \xleftarrow{\$} R_q^{\ell \times \nu}, \mathbf{u} \xleftarrow{\$} R_q^\ell, b \leftarrow \mathbf{A}(\mathbf{A}, \mathbf{u}) \right] \right| \geq \varepsilon.$$

The parameter ℓ does not play a crucial role in determining the hardness of the above problems, so we often omit it. Additionally, if distributions χ and ψ are spherical discrete Gaussian distributions, we replace them with their width parameters for simplicity.

Definition 5 (Hint-MLWE). Let ν, ℓ be positive integers, and $\chi, \psi_1, \dots, \psi_n$ be distributions over R . We say that a PPT adversary $\mathbf{A}$ has advantage ε in solving $\text{HMLWE}_{q,\nu,\ell,\chi,\psi_1,\dots,\psi_n}^{c_1,\dots,c_n}$ for $c_1, \dots, c_n \in R$ if

$$\left| \Pr \left[b = 1 \mid \begin{array}{l} \mathbf{A} \xleftarrow{\$} R_q^{\ell \times \nu}, \mathbf{r} \xleftarrow{\$} \chi^{\nu+\ell}, \mathbf{s}_1 \leftarrow \psi_1^{\nu+\ell}, \dots, \mathbf{s}_n \xleftarrow{\$} \psi_n^{\nu+\ell} \\ b \leftarrow \mathbf{A}(\mathbf{A}, [\mathbf{A} | \mathbf{I}_\ell] \mathbf{r}, \mathbf{r} + c_1 \cdot \mathbf{s}_1, \dots, \mathbf{r} + c_n \cdot \mathbf{s}_n) \end{array} \right] - \Pr \left[b = 1 \mid \begin{array}{l} \mathbf{A} \xleftarrow{\$} R_q^{\ell \times \nu}, \mathbf{r} \xleftarrow{\$} \chi^{\nu+\ell}, \mathbf{s}_1 \xleftarrow{\$} \psi_1^{\nu+\ell}, \dots, \mathbf{s}_n \xleftarrow{\$} \psi_n^{\nu+\ell} \\ \mathbf{u} \xleftarrow{\$} R_q^\ell, b \leftarrow \mathbf{A}(\mathbf{A}, \mathbf{u}, \mathbf{r} + c_1 \cdot \mathbf{s}_1, \dots, \mathbf{r} + c_n \cdot \mathbf{s}_n) \end{array} \right] \right| \geq \varepsilon.$$

If all randomness are sampled from discrete Gaussian distributions, then there exists an efficient reduction from the MLWE problem to the Hint-MLWE problem, which is described as follows.

Theorem 1 (Theorem 1 [KLSS23]). Let $c_1, \dots, c_n \in R$ satisfying $\|c_i\|_1 \leq B_i$ for $1 \leq i \leq n$. Let $\sigma, \sigma_0, \dots, \sigma_n > 0$ satisfy $1/\sigma^2 = 2(1/\sigma_0^2 + \sum_{i=1}^n B_i^2/\sigma_i^2)$ and $\sigma \geq \sqrt{2} \cdot \eta_\varepsilon(\mathbb{Z}^d)$ for some $\varepsilon = \text{negl}(\lambda)$. Then, there exists an efficient reduction from $\text{MLWE}_{q,\nu,\sigma}$ to $\text{HMLWE}_{q,\nu,\sigma_0,\sigma_1,\dots,\sigma_n}^{c_1,\dots,c_n}$.

3.4 Commitment Scheme

Definition 6 (Commitment). A commitment scheme is a tuple of PPT algorithms $(\text{Setup}, \text{Com}, \text{Open})$ with the following syntax.

- $\text{Setup}(1^\lambda) \rightarrow \text{pp}$: Given a security parameter λ , output public parameters pp .
- $\text{Com}(\text{pp}, m) \rightarrow (\text{com}, \text{op})$: Given public parameters pp and a message m , output a commitment com and an opening op .
- $\text{Open}(\text{pp}, \text{com}, m, \text{op}) \rightarrow 0/1$: Given public parameters pp , a commitment com , a message m , and an opening op , output 0 or 1.

Definition 7 (Binding). A commitment scheme $(\text{Setup}, \text{Com}, \text{Open})$ satisfies binding if for every PPT adversary $\mathbf{A}$, the following holds.

$$\Pr \left[\begin{array}{l} m \neq m' \wedge \\ \text{Open}(\text{pp}, \text{com}, m, \text{op}) = 1 \wedge \\ \text{Open}(\text{pp}, \text{com}, m', \text{op}') = 1 \end{array} \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (\text{com}, m, m', \text{op}, \text{op}') \leftarrow \mathbf{A}(\text{pp}) \end{array} \right] = \text{negl}(\lambda)$$

Definition 8 (Hiding). A commitment scheme $(\text{Setup}, \text{Com}, \text{Open})$ satisfies hiding if for every PPT adversary $\mathbf{A}$, the following holds.

$$\left| \Pr \left[b = b' \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda), (m_0, m_1) \leftarrow \mathbf{A}(\text{pp}) \\ b \xleftarrow{\$} \{0, 1\}, (\text{com}, \text{op}) \leftarrow \text{Com}(\text{pp}, m_b) \\ b' \leftarrow \mathbf{A}(\text{com}) \end{array} \right] - \frac{1}{2} \right| \leq \text{negl}(\lambda).$$

4 Proof System with Relaxed Completeness

In this section, we formalize the notions of relaxed argument of knowledge and relaxed reduction of knowledge to address peculiarities in lattice-based proof systems. Specifically, there exists a gap between the statement that the prover intends to prove and what the verifier can actually check in lattice-based proof systems. We address this issue by introducing a relaxation on completeness, which distinguishes between the witness-instance relations that the prover can use for proof generation and those that the verifier can verify. This approach follows the design principle of prior lattice-based proof systems [ALS20, LNP22, BS23]. We then show that the useful properties of the standard reduction of knowledge framework [KP23] are preserved under our generalized definition. Finally, we define the security notion of a PCS based on this generalized framework.

4.1 Relaxed Interactive Argument of Knowledge

Definition 9 (Relaxed Interactive Argument of Knowledge). Let $\Pi = (\text{Setup}, \text{P}, \text{V})$ be an interactive protocol with following syntax.

- $\text{Setup}(1^\lambda) \rightarrow \text{pp}$: Given a security parameter λ , output public parameters pp .
- $\text{P}(\text{pp}, x_1, w_1) \rightarrow \perp$: Given public parameters pp , a public instance x_1 , and a witness w_1 , output nothing after interacting with V .
- $\text{V}(\text{pp}, x_1) \rightarrow b$: Given public parameters pp , a public instance x_1 , output a bit b after interacting with P .

The protocol Π is called a relaxed argument of knowledge for a relation pair (R, R') if $\text{R} \subseteq \text{R}'$ and it satisfies the following properties.

- **Relaxed Completeness:** For all relations $(x, w) \in \text{R}$, the following holds.

$$\Pr \left[b = 1 \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ b \leftarrow \langle \text{P}(\text{pp}, x, w), \text{V}(\text{pp}, x) \rangle \end{array} \right] = 1 - \text{negl}(\lambda)$$

- **Knowledge Soundness:** For all expected PPT adversaries $\mathbf{A}$ and $\mathbf{P}^*$, there exists an expected PPT extractor $\mathbf{E}$ such that the following holds.

$$\Pr \left[b = 1 \wedge (x, w) \notin \text{R}' \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (\text{st}, x) \leftarrow \mathbf{A}(\text{pp}) \\ b \leftarrow \langle \mathbf{P}^*(\text{pp}, \text{st}), \text{V}(\text{pp}, x) \rangle \\ w \leftarrow \mathbf{E}(\text{pp}, \text{st}, x) \end{array} \right] = \text{negl}(\lambda)$$

4.2 Relaxed Reduction of Knowledge

Definition 10 (Relaxed Reduction of Knowledge). Let $\Pi = (\text{Setup}, P, V)$ be an interactive protocol with the following syntax.

- $\text{Setup}(1^\lambda) \rightarrow \text{pp}$: Given a security parameter λ , output public parameters pp .
- $\underline{P}(\text{pp}, x_1, w_1) \rightarrow (x_2, w_2)$: Given public parameters pp , a public instance x_1 , and a witness w_1 , output a reduced instance-witness pair (x_2, w_2) after interacting with V .
- $\underline{V}(\text{pp}, x_1) \rightarrow x_2$: Given public parameters pp , a public instance x_1 , output a reduced instance x_2 after interacting with P .

The protocol Π is called a relaxed reduction of knowledge from a relation pair (R_1, R'_1) to a relation pair (R_2, R'_2) if $R_1 \subseteq R'_1$, $R_2 \subseteq R'_2$, and it satisfies the following properties.

- **Relaxed Completeness:** For all relations $(x_1, w_1) \in R_1$, the following holds.

$$\Pr \left[(x_2, w_2) \in R_2 \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (x_2, w_2) \leftarrow \langle P(\text{pp}, x_1, w_1), V(\text{pp}, x_1) \rangle \end{array} \right] = 1 - \text{negl}(\lambda)$$

- **Knowledge Soundness:** For all expected PPT adversaries A and P^* , there exists an expected PPT extractor E such that the following holds.

$$\Pr \left[\begin{array}{l} (x_2, w_2) \in R'_2 \wedge \\ (x_1, w_1) \notin R'_1 \end{array} \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (\text{st}, x_1) \leftarrow A(\text{pp}) \\ (x_2, w_2) \leftarrow \langle P^*(\text{pp}, \text{st}), V(\text{pp}, x_1) \rangle \\ w_1 \leftarrow E(\text{pp}, \text{st}, x_1) \end{array} \right] = \text{negl}(\lambda)$$

- **Public Reducibility:** For all expected PPT adversaries A and P^* , there exists a deterministic polynomial-time function ϕ that satisfies the following condition.

$$\Pr \left[\phi(\text{pp}, x_1, \text{tr}) = x_2 \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (\text{st}, x_1) \leftarrow A(\text{pp}) \\ (\text{tr}, x_2) \leftarrow \langle P^*(\text{pp}, \text{st}), V(\text{pp}, x_1) \rangle \end{array} \right] = 1$$

Below, we show that the key results from the reduction of knowledge framework hold under our relaxed definition.

Theorem 2. Let Π_1 be a relaxed reduction of knowledge from (R_1, R'_1) to (R_2, R'_2) , and let Π_2 be a relaxed argument of knowledge for the relation pair (R_2, R'_2) . Then, $\Pi_2 \circ \Pi_1$ is a relaxed argument of knowledge for the relation pair (R_1, R'_1) .

Proof. See Section C.1. □

4.3 Polynomial Commitment Scheme

We define the security notion of a PCS based on the definition of a relaxed argument of knowledge, which is analogous to the weak binding definition in [NS24].

Definition 11 (Polynomial Commitment). *A polynomial commitment scheme is a tuple of PPT algorithms $(\text{Setup}, \text{Com}, \text{Open}, \text{Eval})$ with the following syntax.*

- $\text{Setup}(1^\lambda, N) \rightarrow \text{pp}$: Given a security parameter λ and a degree bound N , output public parameters pp .
- $\text{Com}(\text{pp}, f) \rightarrow (\text{com}, \text{op})$: Given public parameters pp and a polynomial $f \in \mathbb{Z}_p^{<N}[X]$, output a commitment com and an opening op .
- $\text{Open}(\text{pp}, \text{com}, f, \text{op}) \rightarrow 0/1$: Given public parameters pp , a commitment com , a polynomial $f \in \mathbb{Z}_p^{<N}[X]$, and an opening op , output 0 or 1.
- $\text{Eval} := (\text{Eval.P}, \text{Eval.V})$ is a relaxed interactive argument of knowledge for a relation pair (R, R') , where R' is defined as follows for an instance $x = (\text{com}, x, y)$ with $x, y \in \mathbb{Z}_p$ and a witness $w = (f, \text{op})$.

$$R' := \left\{ (x, w) \mid \begin{array}{l} \text{Open}(\text{pp}, \text{com}, f, \text{op}) = 1 \wedge \\ f(x) = y \wedge \deg(f) < N \end{array} \right\} \quad (4)$$

Definition 12 (Evaluation Completeness). *A polynomial commitment scheme $(\text{Setup}, \text{Com}, \text{Open}, \text{Eval})$ satisfies evaluation completeness if the following holds for all PPT adversaries A .*

$$\Pr \left[\begin{array}{c} (x, w) \in R \\ x := (\text{com}, x, f(x)), w := (f, \text{op}) \end{array} \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda, N) \\ (f, x) \leftarrow A(\text{pp}) \\ (\text{com}, \text{op}) \leftarrow \text{Com}(\text{pp}, f) \end{array} \right] = 1 - \text{negl}(\lambda)$$

Definition 13 (Evaluation Hiding). *A polynomial commitment scheme $(\text{Setup}, \text{Com}, \text{Open}, \text{Eval})$ satisfies hiding if for every PPT adversary A , there exists a PPT simulator S satisfying the following.*

$$\left| \Pr \left[\begin{array}{c} \deg(f) < N \wedge \\ A(\text{pp}, \text{com}, \text{tr}) = 1 \end{array} \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda, N) \\ (f, x) \leftarrow A(\text{pp}) \\ (\text{com}, \text{op}) \leftarrow \text{Com}(\text{pp}, f) \\ x := (\text{com}, x, f(x)), w := (f, \text{op}) \\ \text{tr} \leftarrow \langle \text{Eval.P}(\text{pp}, x, w), \text{Eval.V}(\text{pp}, x) \rangle \end{array} \right] \right. \\ \left. - \Pr \left[\begin{array}{c} \deg(f) < N \wedge \\ A(\text{pp}, \text{com}, \text{tr}) = 1 \end{array} \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda, N) \\ (f, x) \leftarrow A(\text{pp}) \\ (\text{com}, \text{tr}) \leftarrow S(\text{pp}, x, f(x)) \end{array} \right] \right| = \text{negl}(\lambda)$$

5 Jindo Polynomial Commitment Scheme

In this section, we construct a new lattice-based polynomial commitment scheme (PCS), Jindo, which supports a large base field, a large challenge set, and the evaluation hiding property, overcoming previous limitations in lattice-based PCSs

CELPC [HSS24] and Greyhound [NS24]. We first review the encoding method in CELPC for supporting large base fields. Then, we construct a new polynomial evaluation protocol by integrating techniques from CELPC and Greyhound. Finally, we construct efficient batched evaluation protocols leveraging the homomorphic property of Jindo. We defer all proofs to Section C.

5.1 CELPC Encoding Method

We review the encoding method introduced in CELPC [HSS24], as illustrated in Fig. 1. We assume that the coefficient space is given as $\mathbb{Z}_p$ and the message space is $R = \mathbb{Z}[X]/(X^d + 1)$, and there exists some integer $b > 0$ such that $p = b^d + 1$.

- $\text{Ecd}(a) \rightarrow f$: Given an integer $a \in \mathbb{Z}_p$, return -1 if $a = -1 \pmod{p}$, otherwise return a polynomial $f = \sum_{i=0}^{d-1} a_i X^i \in R = \mathbb{Z}[X]/(X^d + 1)$ where $(a_{d-1} \cdots a_0)_b$ is the base b representation of a .
- $\text{REcd}_\sigma(a) \rightarrow g$: Given an integer $a \in \mathbb{Z}_p$, return a polynomial $g \in R$, whose coefficients are sampled from $\mathcal{D}_{\text{Ecd}(a) + \mathbf{T}\mathbb{Z}^d, \sigma\mathbf{T}}$, where $\mathbf{T}$ is the negacyclic matrix of $X - b$.
- $\text{Dcd}(f) \rightarrow a$: Given a polynomial $f \in R$, return $a = f(b) \pmod{p}$.

Fig. 1: Encoding Method of Jindo

The key advantage of the CELPC encoding method is that it effectively reduces the norm of messages by $b \ll p$ when committing polynomials in $\mathbb{Z}_p[X]$, while maintaining the homomorphic property. Below, we summarize norm bounds for the encoding method and its homomorphic property.

Lemma 2 (Norm Bounds). *The following hold for all $a \in \mathbb{Z}_p$ and $\sigma > \eta_{1/3}(\mathbb{Z}^d)$.*

- $\|\text{Ecd}(a)\|_\infty \leq b$
- $\Pr \left[\|\text{REcd}_\sigma(a)\|_\infty \leq 5\sigma(b+1) \right] \geq 1 - d \cdot 2^{-111}$

Proof. See Lemma 8. □

Lemma 3 (Homomorphism). *$\text{Dcd} : R \rightarrow \mathbb{Z}_p$ is a ring homomorphism, and $(\text{Dcd} \circ \text{REcd}_\sigma)(a) = (\text{Dcd} \circ \text{Ecd})(a) = a$ holds for all $a \in \mathbb{Z}_p$.*

Proof. See Lemma 9 in [HSS24]. □

5.2 Scheme Description

We describe the PCS algorithms of Jindo in Fig. 2 and provide a security analysis as a commitment scheme.

- **Setup**($1^\lambda, N$) $\rightarrow$ **pp**: Given a security parameter λ and a degree upper bound N , return public parameters **pp** defined as follows.
 - $q, q_h, q_o, q_s \in \mathbb{Z}_{>0}$.
 - $N = mn$, $R = \mathbb{Z}[X]/(X^d + 1)$, and $p = b^d + 1$.
 - $\mathbf{A} \xleftarrow{\$} R_q^{\mu \times (m+1)}$, $\mathbf{B} := [\mathbf{B}' \mid \mathbf{I}_\mu] \in R_q^{\mu \times (\mu + \nu)}$ for $\mathbf{B}' \xleftarrow{\$} R_q^{\mu \times \nu}$, and $\mathbf{D} \xleftarrow{\$} R_{q_o}^{\kappa \times \mu(n+1)}$.
- **Com**(**pp**, f) $\rightarrow$ (**û**, **op**): Given a polynomial $f(X) = \sum_{i=0}^{N-1} f_i X^i$, return a commitment $\hat{\mathbf{u}} \in R^\kappa$ and an opening **op** = ($\hat{\mathbf{F}}, \mathbf{R}, \hat{\mathbf{t}}, \mathbf{1}, 1$), which are computed as follows.
 1. Sample $b_1, \dots, b_{n-1} \xleftarrow{\$} \mathbb{Z}_p$, and define $f_{i,j} \in \mathbb{Z}_p$ as follows.
 - $(f_{0,0}, \dots, f_{0,n-1}) := (f_0, f_1 - b_1, \dots, f_{n-1} - b_{n-1})$.
 - $(f_{i,0}, \dots, f_{i,n-1}) := (f_{in}, \dots, f_{(i+1)n-1})$ for $1 \leq i < n$.
 - $(f_{m,0}, \dots, f_{m,n-1}) := (b_1, \dots, b_{n-1}, 0)$.
 - $f_{0,n}, \dots, f_{m,n} \xleftarrow{\$} \mathbb{Z}_p$.
 2. Sample $\hat{\mathbf{f}}_j = (f_{0,j}, \dots, f_{m,j}) \in R^{m+1}$ as follows, and define $\hat{\mathbf{F}} := [\hat{\mathbf{f}}_0 \mid \dots \mid \hat{\mathbf{f}}_n] \in R^{(m+1) \times (n+1)}$.
 - $\hat{f}_{0,j} \xleftarrow{\$} \text{REcd}_{\sqrt{m+1} \cdot \sigma_0}(f_{0,j})$ for $0 \leq j < n$.
 - $\hat{f}_{i,j} \xleftarrow{\$} \text{REcd}_{\sigma_1}(f_{i,j})$ for $1 \leq i \leq m$ and $0 \leq j < n$.
 - $\hat{f}_{0,n} \xleftarrow{\$} \text{REcd}_{\sqrt{mn+1} \cdot \sigma_2}(f_{0,n})$.
 - $\hat{f}_{i,n} \xleftarrow{\$} \text{REcd}_{\sqrt{n+1} \cdot \sigma_3}(f_{i,n})$ for $1 \leq i \leq m$.
 3. Sample $\mathbf{r}_j \in R^{\mu+\nu}$ as follows, and let $\mathbf{R} := [\mathbf{r}_0 \mid \dots \mid \mathbf{r}_n] \in R^{(\mu+\nu) \times (n+1)}$.
 - $\mathbf{r}_j \xleftarrow{\$} \mathcal{D}_{\mathbb{Z}^d, \tau_0}^{\mu+\nu}$ for $0 \leq j < n$.
 - $\mathbf{r}_n \xleftarrow{\$} \mathcal{D}_{\mathbb{Z}^d, \sqrt{n+1} \cdot \tau_1}^{\mu+\nu}$.
 4. Compute $\mathbf{t}_j = \mathbf{A}\hat{\mathbf{f}}_j + \mathbf{B}\mathbf{r}_j \in R_q^\mu$ and $\hat{\mathbf{t}}_j = \lfloor \mathbf{t}_j / q_h \rfloor \in R^\mu$ for $0 \leq j \leq n$, and define $\hat{\mathbf{t}} := \hat{\mathbf{t}}_0 \parallel \dots \parallel \hat{\mathbf{t}}_n \in R^{\mu(n+1)}$.
 5. Compute $\mathbf{u} = \mathbf{D}\hat{\mathbf{t}} \in R_{q_o}^\kappa$, and let $\hat{\mathbf{u}} := \lfloor \mathbf{u} / q_s \rfloor \in R^\kappa$.
- **Open**(**pp**, f , $\hat{\mathbf{u}}$, **op**) $\rightarrow$ 0/1: Given a polynomial $f \in \mathbb{Z}_p^{<N}[X]$, a commitment $\mathbf{u} \in R_q^\kappa$, and an opening **op** = ($\hat{\mathbf{F}}, \mathbf{R}, \hat{\mathbf{t}}, \mathbf{c}, \alpha$), where $\hat{\mathbf{F}} = [\hat{\mathbf{f}}_0 \mid \dots \mid \hat{\mathbf{f}}_n]$, $\hat{\mathbf{t}} = \hat{\mathbf{t}}_0 \parallel \dots \parallel \hat{\mathbf{t}}_n$, $\mathbf{R} = [\mathbf{r}_0 \mid \dots \mid \mathbf{r}_n]$, and $\mathbf{c} = (c_0, \dots, c_{n-1})$, output 1 if the following conditions hold.
 1. Check $\|\alpha\|_1 \leq B_\alpha$ and $\|c_j\|_1 \leq B_c$ for $0 \leq j < n$.
 2. Define $\mathbf{s} := \mathbf{D}\hat{\mathbf{t}} - q_s \alpha \cdot \hat{\mathbf{u}} \in R_q^\mu$.
 3. Check $\|\mathbf{s}\|_2 + \|\hat{\mathbf{t}}\|_2 \leq B_o$, and $\|\alpha\|_1 \leq B_\alpha$.
 4. Define $\mathbf{h}_j := \mathbf{A}\hat{\mathbf{f}}_j + \mathbf{B}\mathbf{r}_j - q_h c_j \cdot \hat{\mathbf{t}}_j \in R_q^\mu$ for $0 \leq j \leq n$.
 5. Check $\|\hat{\mathbf{f}}_j\|_2 + \|\mathbf{r}_j\|_2 + \|\mathbf{h}_j\|_2 \leq B_j$ for $0 \leq j \leq n$.
 6. Define $f_{i,j} \in \mathbb{Z}_p$ as follows for $\hat{\mathbf{f}}_j = (\hat{f}_{0,j}, \dots, \hat{f}_{m,j})$ and $0 \leq j < n$.

$$f_{i,j} := \text{Dcd}(\alpha c_j)^{-1} \cdot \text{Dcd}(\hat{f}_{i,j}) \in \mathbb{Z}_p$$
 7. Check that the following holds.

$$f(X) = [1 \ X^n \ \dots \ X^{N-n} \ X] \begin{bmatrix} f_{0,0} & \dots & f_{0,n-1} \\ \vdots & & \vdots \\ f_{m,0} & \dots & f_{m,n-1} \end{bmatrix} \begin{bmatrix} 1 \\ X \\ \vdots \\ X^{n-1} \end{bmatrix}$$

– $(\text{Eval.P}(\text{pp}, \mathbf{x}, \mathbf{w}), \text{Eval.V}(\text{pp}, \mathbf{x}))$: Run the interactive protocol in Fig. 3 for an instance $\mathbf{x} = (\mathbf{u}, x, y)$ and a witness $\mathbf{w} = (f, \text{op})$, where $\hat{\mathbf{x}} := (1, \text{Ecd}(x^n), \dots, \text{Ecd}(x^{N-n}), \text{Ecd}(x)) \in R^{m+1}$, and $\tilde{\mathbf{x}} := (1, \text{Ecd}(x), \dots, \text{Ecd}(x^{n-1}), 0) \in R^{n+1}$.

Fig. 2: Polynomial Commitment Algorithms of Jindo

For the commitment algorithm **Com**, we employ randomized encoding from [HSS24] and commitment compression from [DKL⁺18, ENS20, LNS20]. The randomized encoding technique is not required for the hiding property of the commitment scheme itself, but is crucial for achieving evaluation hiding in the evaluation protocol, which we analyze later. The commitment compression technique is applied to both the inner commitments $\mathbf{t}_j$ and the outer commitment $\mathbf{u}$, by replacing them with rounded values $\hat{\mathbf{t}}_j := \lfloor \mathbf{t}_j / q_h \rfloor$ and $\hat{\mathbf{u}} := \lfloor \mathbf{u} / q_s \rfloor$. This not only reduces the overhead of outer commitment generation but also reduces communication costs in the evaluation protocol by decreasing the size of both the inner and outer commitments.

For the opening algorithm **Open**, we follow the weak opening framework developed in lattice-based proof systems [BDL⁺18, BS23, NS24], which allows relaxation factors $\mathbf{c}$ and α in the opening process to handle the inherent gap between the provable statement and the verifiable statement in lattice-based proof systems. Although we allow these relaxation factors, we can still ensure that the commitment scheme maintains the binding property for the committed message.

Security Analysis. The binding and hiding properties of Jindo follow from the MSIS and MLWE problems, respectively, as described below.

Theorem 3 (Binding). *The commitment scheme defined in Fig. 2 satisfies binding if $\text{MSIS}_{q_o, \kappa, 2B_\alpha B_o}$ and $\text{MSIS}_{q, \mu, 2B_\alpha B_c B_j}$ are computationally hard for $0 \leq j < n$.*

Theorem 4 (Hiding). *The commitment scheme defined in Fig. 2 satisfies hiding if both $\text{MLWE}_{q, \nu, \tau_0}$ and $\text{MLWE}_{q, \nu, \sqrt{n+1} \cdot \tau_1}$ are computationally hard.*

5.3 Evaluation Protocol

We present the polynomial evaluation protocol of Jindo in Fig. 3. Below, we provide its security and complexity analysis.

Security Analysis. We prove that the evaluation protocol Π_{Eval} in Fig. 3 is a relaxed argument of knowledge protocol and satisfies evaluation completeness and evaluation hiding properties. We first define the following three witness–instance relations to describe the relations supported by our protocol.

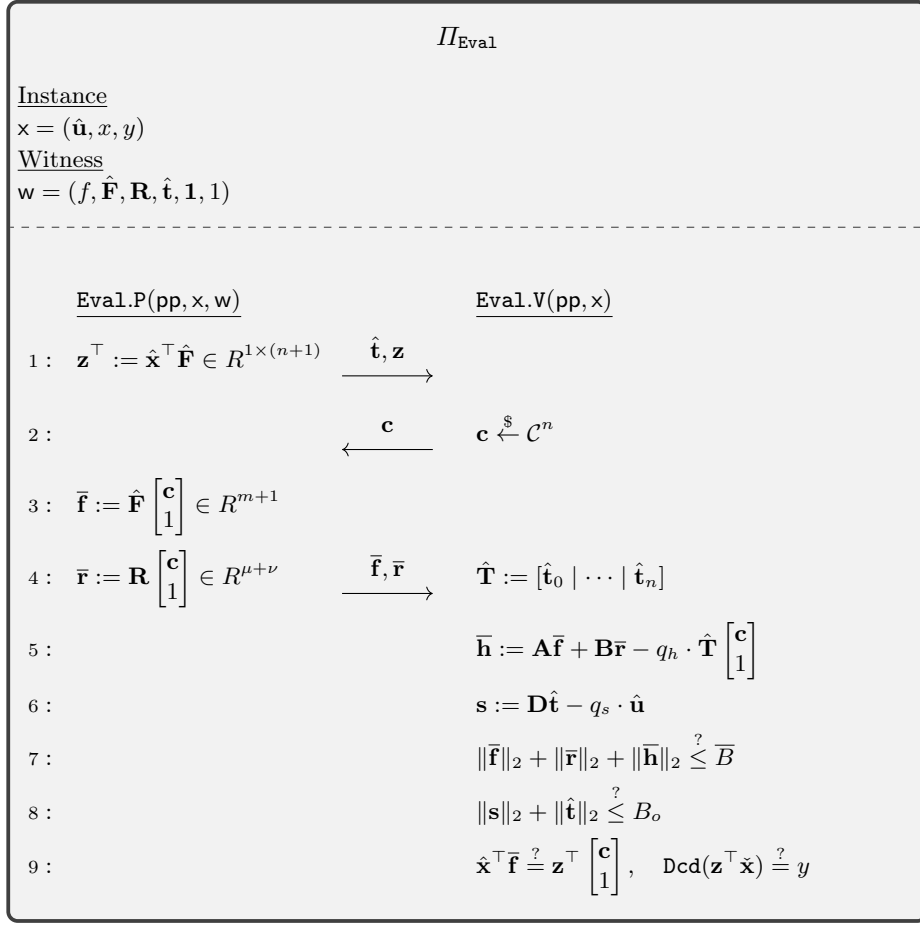


Fig. 3: Evaluation Protocol of Jindo

$$\begin{aligned}
- R_{\text{PCS}}^{(B_j, B_o, B_c, B_\alpha)} &:= \left\{ (x, w) \left| \begin{array}{l} f(x) = y \wedge \deg(f) < N \wedge \\ \text{Open}(\text{pp}, \mathbf{u}, f, \text{op}) = 1 \end{array} \right. \right\} \\
- R_{\text{PCS}}^{(B_j, B_o, B_c)} &:= \left\{ (x, w) \left| \alpha = 1 \wedge (x, w) \in R_{\text{PCS}}^{(B_j, B_o, B_c, 1)} \right. \right\} \\
- R_{\text{PCS}}^{(B_j, B_o)} &:= \left\{ (x, w) \left| \mathbf{c} = \mathbf{1} \wedge (x, w) \in R_{\text{PCS}}^{(B_j, B_o, 1)} \right. \right\}
\end{aligned}$$

Fig. 4: Relations for the Jindo PCS

The first relation is the most general form, where we consider relaxation factors for both the outer and inner commitments. This relation is used to prove the

knowledge soundness of the batched evaluation protocol later. The second relation considers only the relaxation factor associated with the inner commitments, corresponding to the case where an extractor obtains a witness through rewinding in the evaluation protocol when proving knowledge soundness. Finally, the last relation corresponds to the plain case, where the prover honestly generates commitments using the commitment algorithm Com , which we use to prove the completeness of the protocol. Based on these relations, we prove that Π_{Eval} is a relaxed argument of knowledge below.

Theorem 5 (Relaxed Argument of Knowledge). *Let $\mathcal{C} \subseteq R$ such that $1/|\text{Dcd}(\mathcal{C})| = \text{negl}(\lambda)$, and let $B_{\mathcal{C}} > 0$ such that $\|c\|_1 \leq B_{\mathcal{C}}$ for $c \xleftarrow{\$} \mathcal{C}$. Then, the interactive protocol $(\text{Eval.P}, \text{Eval.V})$ defined in Fig. 3 is a relaxed interactive argument of knowledge for $(R_{\text{PCS}}^{(B_j, B_o)}, R_{\text{PCS}}^{(B'_j, B'_o, B'_c)})$, where the norm bounds are defined as $\bar{B} = \sum_{j=0}^{n-1} B_{\mathcal{C}} B_j + B_n$, $B'_j := 2\bar{B}$, $B'_o := B_o$, and $B'_c := 2B_{\mathcal{C}}$.*

Next, we prove the evaluation completeness of Π_{Eval} , which specifies norm bounds for the relations defined in Fig. 4 when commitments are generated using the algorithm Com in Fig. 2.

Theorem 6 (Evaluation Completeness). *The polynomial commitment scheme defined in Fig. 2 satisfies evaluation completeness with respect to the relation $R_{\text{PCS}}^{(B_j, B_o)}$ for the following norm bounds if $\sigma_0, \sigma_1, \sigma_2, \sigma_3, \tau_0, \tau_1 \geq \eta_\varepsilon(\mathbb{Z}^d)$ for some $0 < \varepsilon < 1/3$.*

- $B_j = 5(b+1)\sqrt{(\sigma_0^2 + \sigma_1^2)(m+1)d} + \tau_0\sqrt{(\mu+\nu)d} + q_h\sqrt{\mu d}$ for $0 \leq j < n$
- $B_n = 5(b+1)\sqrt{(\sigma_2^2 + \sigma_3^2)(m+1)(n+1)d} + \tau_1\sqrt{(n+1)(\mu+\nu)d} + q_h\sqrt{\mu d}$
- $B_o = q_s\sqrt{\kappa d} + (q/q_h)\sqrt{\mu nd}$

Finally, we prove the evaluation hiding property, assuming commitments are generated through Com in Fig. 2. The core idea is similar to CELPC, achieving transcript simulatability by leveraging the properties of discrete Gaussian distributions via the convolution lemma in [Pei10]. To achieve this, we apply randomized encoding to the input messages as described in the commitment algorithm Com in Fig. 2, following the approach in CELPC. This approach results in faster proof generation and simplified protocol analysis compared to the rejection sampling-based method used in [BDL⁺18, LNS20, ENS20, LNP22], a conventional technique for achieving zero-knowledge in lattice-based proof systems that requires frequent re-execution of the proof generation algorithm.

Theorem 7 (Evaluation Hiding). *Let $B_x > 0$ such that $\|\text{Ecd}(x^i)\|_1 \leq B_x$ for $0 \leq i < N$. The polynomial commitment scheme defined in Fig. 2 satisfies hiding if $\text{MLWE}_{q, \nu, \tau}$ is computationally hard for $\tau^{-2} = 2(\tau_0^{-2} + B_{\mathcal{C}}^2 \tau_1^{-2})$ and $\sigma_0, \sigma_1, \sigma_2, \sigma_3, \tau_0, \tau_1$ satisfy the following for some $\varepsilon = \text{negl}(\lambda)$.*

- $\sigma_0 \geq \frac{2B_x}{b-1} \cdot \eta_\varepsilon(\mathbb{T}\mathbb{Z}^d)$, $\sigma_1 \geq \frac{2}{b-1} \cdot \eta_\varepsilon(\mathbb{T}\mathbb{Z}^d)$.
- $\sigma_2 \geq \frac{2B_{\mathcal{C}}B_x}{b-1} \cdot \eta_\varepsilon(\mathbb{T}\mathbb{Z}^d)$, $\sigma_3 \geq \frac{2B_{\mathcal{C}}}{b-1} \cdot \eta_\varepsilon(\mathbb{T}\mathbb{Z}^d)$.
- $\tau_0 \geq 2\sqrt{2} \cdot \eta_\varepsilon(\mathbb{Z}^d)$, $\tau_1 \geq 2\sqrt{2}B_{\mathcal{C}} \cdot \eta_\varepsilon(\mathbb{Z}^d)$.

Complexity Analysis. We provide an asymptotic complexity analysis of the prover, verifier, and communication cost of Π_{Eval} , which yields results similar to those of the previous constructions CELPC and Greyhound. Specifically, the verification and communication costs follow almost square-root complexity with respect to the degree bound N . Below, we summarize the results.

Theorem 8. *The polynomial commitment scheme defined in Fig. 2 and the evaluation protocol defined in Fig. 3 achieves the following complexity.*

- *Prover complexity:* $O(N \log^2 N)$
- *Verifier complexity:* $O(\sqrt{N} \log^3 N)$
- *Communication complexity:* $O(\sqrt{N} \log N)$

5.4 Batched Evaluation Protocol

While both CELPC and Greyhound address batched evaluation, the efficacy of their solutions is limited. Specifically, their protocols only amortize the costs associated with the final response $\bar{\mathbf{f}}$, while incurring a linear blow-up in verification and communication costs for the inner commitments $\hat{\mathbf{t}}$, whose size is $O(\sqrt{N} \log N)$. In the case of CELPC, this inefficiency stems from the lack of an outer commitment technique that effectively reduces the actual commitment size. For Greyhound, it arises from the absence of a homomorphic encoding method for polynomial coefficients, which prevents homomorphic linear operations directly at the commitment level.

In contrast, we observe that Jindo satisfies all criteria for efficient batched evaluation. By leveraging the outer commitment technique, the commitment size is reduced to $O(\log^2 N)$, which is significantly smaller than the inner commitments or final responses, which scale as $O(\sqrt{N} \log N)$. Additionally, Jindo inherits linear homomorphism from the CELPC encoding method. Consequently, we can adapt the random linear combination technique from [BDFG21] to the lattice setting, which confines the linear cost blow-up to the commitment level, while leaving the cost of the polynomial evaluation protocol unchanged. We present the full protocol in Fig. 5.

Security Analysis. We analyze the security of the batched evaluation protocol Π_{Batch} through the relaxed reduction of knowledge framework, which we introduce in Section 4.2. Below, we prove that the protocol Π_{Batch} in Fig. 5 is a relaxed reduction of knowledge protocol that reduces ℓ PCS relations into a single PCS relation.

Theorem 9. *Let $\mathcal{C} \subseteq R$ such that $1/|\text{Dcd}(\mathcal{C})| = \text{negl}(\lambda)$, and let $B_{\mathcal{C}} > 0$ such that $\|c\|_1 \leq B_{\mathcal{C}}$ for $c \xleftarrow{\$} \mathcal{C}$. Then, an interactive protocol $(\text{Batch.P}, \text{Batch.V})$ defined in Fig. 5 is a relaxed reduction of knowledge from $\left(\prod_{k=0}^{\ell-1} \mathbf{R}_{\text{PCS}}^{(B_j, B_o)}, \prod_{k=0}^{\ell-1} \mathbf{R}_{\text{PCS}}^{(B'_j, B'_o, B'_c, B'_\alpha)}\right)$ to $\left(\mathbf{R}_{\text{PCS}}^{(\underline{B}_j, \underline{B}_o)}, \mathbf{R}_{\text{PCS}}^{(\underline{B}'_j, \underline{B}'_o, \underline{B}'_c)}\right)$, where the norm bounds are defined as follows.*

- $\underline{B}_j = \ell B_{\mathcal{C}} B_j$, $\underline{B}_o = \ell B_{\mathcal{C}} B_o$.

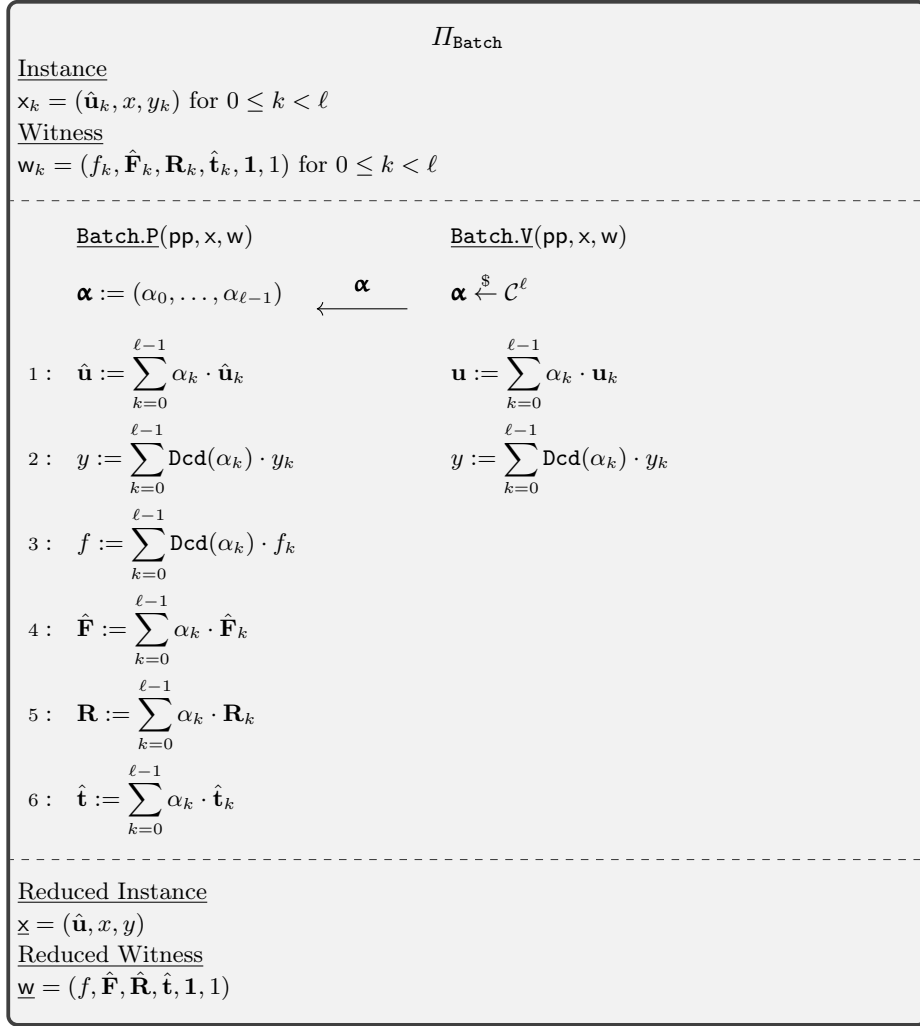


Fig. 5: Reduction of Knowledge Protocol for Batched Evaluation

$$- B'_j = 2\underline{B}'_c \underline{B}'_j, \quad B'_o = 2\underline{B}'_o, \quad B'_c = (\underline{B}'_c)^2, \quad B'_\alpha = 2B_c$$

As a corollary, we can construct an efficient polynomial evaluation protocol that can handle multiple polynomials at once by composing the batched evaluation protocol with the polynomial evaluation protocol, as described below.

Corollary 1. *Suppose the interactive protocol Π_{Batch} defined in Fig. 5 is a relaxed reduction of knowledge from $\left(\prod_{k=0}^{\ell-1} \mathbf{R}_{\text{PCS}}^{(B_j, B_o)}, \prod_{k=0}^{\ell-1} \mathbf{R}_{\text{PCS}}^{(B'_j, B'_o, B'_c, B'_\alpha)} \right)$ to $\left(\mathbf{R}_{\text{PCS}}^{(\underline{B}_j, \underline{B}_o)}, \mathbf{R}_{\text{PCS}}^{(\underline{B}'_j, \underline{B}'_o, \underline{B}'_c)} \right)$, and the interactive protocol Π_{Eval} defined in Fig. 3 is*

a relaxed argument of knowledge for $(R_{\text{PCS}}^{(B_j, B_o)}, R_{\text{PCS}}^{(B'_j, B'_o, B'_c)})$. Then, $\Pi_{\text{BatchEval}} := \Pi_{\text{Eval}} \circ \Pi_{\text{Batch}}$ is a relaxed argument of knowledge for $(\prod_{k=0}^{\ell-1} R_{\text{PCS}}^{(B_j, B_o)}, \prod_{k=0}^{\ell-1} R_{\text{PCS}}^{(B'_j, B'_o, B'_c, B'_\alpha)})$, where the following norm bounds hold.

- $B'_j = 8\ell B_C^2 (\sum_{j=0}^{n-1} B_C B_j + B_n)$, $B'_o = 4\ell B_C B_o$,
- $B'_c = 4B_C^2$, $B'_\alpha = 2B_C$.

Complexity Analysis. We analyze the prover, verification, and communication complexity of the composed protocol $\Pi_{\text{BatchEval}} = \Pi_{\text{Eval}} \circ \Pi_{\text{Batch}}$ defined in Corollary 1. Recall that the primary aim of the batched evaluation protocol is to effectively reduce the verification and proof size overhead associated with the number of polynomials, ℓ . Thanks to commitment-level batching, the ℓ -dependent costs grow only linearly with the commitment size $O(\log^2 N)$, while the cost of the underlying evaluation protocol remains unchanged. Below, we provide a detailed complexity analysis of our protocol.

Theorem 10. *The composed protocol $\Pi_{\text{BatchEval}} = \Pi_{\text{Eval}} \circ \Pi_{\text{Batch}}$ defined in Corollary 1 achieves the following complexity.*

- *Prover complexity:* $O(\ell N \log^2(\ell N))$
- *Verifier complexity:* $O(\sqrt{N} \log^3(\ell N) + \ell \log^2(\ell N))$
- *Communication complexity:* $O(\sqrt{N} \log(\ell N) + \ell \log^2(\ell N))$

6 Implementation

We implement our PCS, Jindo, to demonstrate its practical efficiency. We begin by detailing the parameter selection strategy and optimization techniques deployed in our implementation. Next, we benchmark single polynomial evaluation to quantify the improvements yielded by our enhanced evaluation protocol. We then measure the performance of batched polynomial evaluation to demonstrate the performance gains achieved by our batched evaluation protocol. Finally, we assess Jindo’s performance within PIOP compilations to showcase its comprehensive benefits for constructing PIOP-based ZKAoKs. All benchmarks were conducted in a single-threaded setting on a laptop machine equipped with an Intel i5-11300H CPU running at 3.10 GHz and 16 GB of RAM. Our source code is available at <https://github.com/SNUCP/jindo>.

6.1 Parameter Settings and Optimizations

Below, we present the parameter setting method and optimization techniques employed in our implementation. We also provide a script in our source code that automatically configures the best-performing parameters, given the choice of coefficient space $\mathbb{Z}_p$ and polynomial ring degree N .

Ring Arithmetic. Our PCS requires polynomial ring arithmetic over $R_q = \mathbb{Z}_q[X]/(X^d + 1)$ and $R_{q_o} = \mathbb{Z}_{q_o}[X]/(X^d + 1)$ for the generation of inner and

outer commitments, respectively. To implement ring arithmetic efficiently, it is common to select a prime modulus q such that $2d \mid q - 1$, enabling fast polynomial multiplication over R_q via the number theoretic transform (NTT). However, some lattice-based proof systems, such as Lazer [LSS24] and Greyhound [NS24], do not support NTT-friendly moduli, requiring additional overhead to handle ring arithmetic. In contrast, our PCS imposes no such constraints on the commitment moduli. Therefore, we can utilize NTT-friendly moduli for both q and q_o , eliminating the overhead associated with constrained moduli choices.

CELPC Encoding. We recall that in Section 5.1, the parameters b and d are configured to satisfy $p = b^d + 1$ when encoding polynomial coefficients in the prime field $\mathbb{Z}_p$, so that we obtain an isomorphism $R/(X - b) \cong \mathbb{Z}_p$. However, having a larger ring degree d often yields better performance, as it reduces the overhead of matrix-vector multiplication in commitment generation. Thus, we utilize the more general form of the CELPC encoding from [HSS24], which breaks this dependency. To be precise, it can be extended to encode vectors in $\mathbb{Z}_p^{d/\gamma}$, based on the isomorphism $R/(X^{d/\gamma} - b) \cong \mathbb{Z}_p^{d/\gamma}$, when $p = b^\gamma + 1$. Then, it allows us to choose a larger ring degree d , as it becomes independent of the choice of b and p . We note that both the evaluation and batched evaluation protocols can be naturally generalized to this setting.

Discrete Gaussian. Since the evaluation hiding property of Jindo heavily depends on the randomized encoding method, efficient sampling of discrete Gaussian distribution is crucial for performance. In particular, we need samples from varying coset with fixed standard deviation. To implement this, we utilize two different samplers depending on the size of the standard deviation. For small values, we use the Twin-CDT sampler [AMR18] which is based on precomputed cumulative distribution tables. Since the memory usage of the Twin-CDT sampler does not scale well, for large deviations we switch to the COSAC sampler [ZSS20].

MSIS and MLWE. The security of our PCS primarily relies on the computational hardness of the MSIS and MLWE problems. To assess the hardness of the MSIS problem, we utilize the heuristic from [MR09, GN08] with a root Hermite factor of $\delta \approx 1.005$, following the analysis in previous works [HSS24, ENS20, LNS20]. For the hardness of the MLWE problem, we employ the estimation techniques provided in [APS15]. Based on these analyses, all MSIS and MLWE parameters are configured to provide 128-bit security.

6.2 Benchmark for Single Polynomial Evaluation

We measure the performance of single polynomial evaluation in Jindo. Specifically, we compare the performance between CELPC to depict the performance improvement over previous constructions. We do not include performance of Greyhound, as its reference implementation does not provide the evaluation hiding property and support for large prime fields, which we aim to achieve. Beyond lattice-based PCSs, we also provide a performance comparison with the recent hash-based PCS DeepFold [GLH⁺25], which also supports a large base field and

evaluation hiding property, similar to our scheme. We configure the base field size as $\log p \approx 255$ for all PCS, and use the reference implementations presented in each paper. For **DeepFold**, we measure the performance of the commitment algorithm and the batched evaluation algorithm for $2N$, following the overhead analysis for zero-knowledge, presented in Section 5.3 of [GLH⁺25]. The benchmark results are presented in Table 2.

N		2^{13}	2^{15}	2^{17}	2^{19}
Commit (s)	CELPC	0.03	0.10	0.35	1.30
	DeepFold	0.11	0.49	2.08	9.17
	Jindo	0.04	0.11	0.39	1.43
Eval (s)	CELPC	0.06	0.11	0.26	0.69
	DeepFold	0.06	0.22	0.89	3.89
	Jindo	0.002	0.006	0.024	0.093
Verify (ms)	CELPC	20	34	62	120
	DeepFold	1.2	1.5	1.9	2.2
	Jindo	5.4	9.2	17.0	32.8
Proof (MB)	CELPC	1.13	1.81	3.22	6.07
	DeepFold	0.17	0.22	0.30	0.37
	Jindo	0.18	0.33	0.64	1.25

Table 2: Benchmark Results of Post-quantum PCSs for $\log p \approx 255$

When comparing **Jindo** with **CELPC**, **Jindo** exhibits slightly slower commitment generation due to the additional requirement of generating outer commitments alongside inner commitments. However, in the evaluation phase, **Jindo** significantly outperforms **CELPC** by leveraging a larger challenge set to eliminate unnecessary repetitions. Consequently, **Jindo** achieves an approximate $1.3\times$ improvement in total proof generation at $N = 2^{19}$ by effectively reducing the cost of the evaluation protocol. Regarding verification and proof size, while both schemes exhibit square-root asymptotic growth, **Jindo** achieves a $3\times$ to $4\times$ improvement in verification speed and a $5\times$ to $6\times$ reduction in proof size, primarily due to the larger challenge space.

Compared to **DeepFold**, **Jindo** achieves significantly faster proof generation in both the commitment and evaluation phases, driven by efficient polynomial ring arithmetic. For instance, **Jindo** achieves an approximate $8.6\times$ speedup in proof generation over **DeepFold** when $N = 2^{19}$. In terms of verification and proof size, **DeepFold** exhibits superior performance due to its polylogarithmic complexity scaling. Consequently, we expect that **Jindo** is the optimal choice for scenarios where the prover’s computational power is restricted but high network bandwidth is available, such as in typical client-server protocols.

6.3 Benchmark for Batched Polynomial Evaluation

To demonstrate the efficacy of batching via linear combination, we measure the performance of the batched evaluation protocol $\Pi_{\text{BatchEval}} = \Pi_{\text{Eval}} \circ \Pi_{\text{Batch}}$. Specifically, we measure the performance of the batched evaluation protocol alongside a plain setting, which simply applies the single polynomial evaluation protocol Π_{Eval} ℓ times for ℓ polynomials. The benchmark results are provided in Table 3.

ℓ	Prover (s)		Verifier (s)		Proof (MB)	
	Plain	Batched	Plain	Batched	Plain	Batched
8	12.26	13.55	0.29	0.08	10.07	2.83
16	24.58	26.40	0.58	0.08	20.14	2.90
32	49.05	53.76	1.17	0.09	40.28	3.23

Table 3: Benchmark Results of Batched Evaluation for $\log p \approx 255$, $N = 2^{19}$

The benchmark results show that the batched evaluation protocol incurs slightly slower proof generation speed compared to the plain setting due to the additional overhead on the prover side, such as the linear combination of witnesses as described in Fig. 5. However, the main goal of batched evaluation is to effectively reduce verification cost and proof size, and our batched evaluation protocol achieves about a $13\times$ reduction in verification cost and a $12.5\times$ reduction in proof size when evaluating 32 polynomials. Specifically, we note that the cost growth in both verification and proof size is much smaller than in the plain setting, since the additional cost from batching grows only linearly with $O(\log^2 N)$, as analyzed in Theorem 10, thanks to the linear homomorphism in Jindo. As a result, our batched evaluation protocol achieves significant cost reduction in verification and proof size when evaluating multiple polynomials simultaneously.

6.4 Benchmark for PIOP Compilations

To demonstrate the efficacy of Jindo in ZKAoK constructions, we measure the performance of it within PIOP compilation scenarios. For the benchmark, we measure the performance of ZKAoKs for the validity of RLWE samples based on the PIOP construction in [HLSS25]. To assess the improvements from our PCS, we provide the performance of ZKAoKs when compiled with Jindo and CELPC. Beyond the PIOP-based approach, we also provide benchmark results from Lazer [LSS24], which also constructs ZKAoKs for RLWE samples based on the BDLOP commitment scheme [BDL⁺18]. In the benchmark, we prove the validity of an RLWE sample $(a, b) \in \mathbb{R}_q = \mathbb{Z}_q[X]/(X^d + 1)$, that it satisfies the relation $b = as + e$ for some $s, e \in \mathbb{R}_q$ with ternary coefficients. We measure

the performance by varying the parameters $\log q$ and d . For **Lazer**, we did not include performance beyond $\log q \approx 440$, and for **CELPC**, performance below $\log q \approx 220$, as their reference implementations do not support this parameter regime. We provide the benchmark results in Table 4.

$(\log q, d)$		$(110, 2^{12})$	$(220, 2^{13})$	$(440, 2^{14})$	$(880, 2^{15})$
Prover (s)	Lazer	1.18	5.77	-	-
	CELPC	-	-	9.33	36.12
	Jindo	0.41	0.85	2.60	17.95
Verify (s)	Lazer	0.62	3.06	-	-
	CELPC	-	-	0.97	2.34
	Jindo	0.03	0.04	0.10	1.40
Proof (MB)	Lazer	0.07	0.14	-	-
	CELPC	-	-	9.1	17.2
	Jindo	0.69	0.65	0.74	1.15

Table 4: Benchmark Results for Proof of RLWE Samples

When compared with **CELPC**, **Jindo** improves performance of ZKAoK across all areas. We attribute this improvement to the enhanced evaluation protocol and efficient batched evaluation. Specifically, **Jindo** achieves an order of magnitude improvement in proof size, thanks to the efficient batching technique enabled by linear homomorphism. For comparison with **Lazer**, **Jindo** shows superior performance in proof generation speed and verification speed, while yielding a larger proof size. For the faster proof generation, direct comparison is challenging due to differing implementation choices, but we attribute the speedup primarily to the elimination of rejection sampling and the use of NTT-friendly moduli. In the case of verification, we note that commitment verification in **Jindo** scales with square-root growth due to the use of the Ajtai commitment, while in **Lazer** it grows linearly due to the BDLOP commitment. This results in more than an order of magnitude speedup in verification.

7 Conclusion

In this work, we construct a new lattice-based PCS resolving the limitations in the previous constructions **CELPC** and **Greyhound**. Our PCS is readily applicable for compilation from PIOP to ZKAoK by supporting a large base field, the evaluation hiding property, and efficient batched evaluation. Particularly, our PCS yields fast proof generation with reasonable proof size and verification speed, making it suitable for settings where the prover has limited computing power and is connected to the verifier via a high-bandwidth network, which is typical

in client-server-based privacy-preserving protocols. As a future work, our PCS can be naturally generalized to support multilinear polynomials following the evaluation method in [WTS⁺18], broadening its applicability for various multilinear PIOPs [CBBZ23]. Additionally, the proof size of our PCS can be further reduced to polylogarithmic scale via composition with Labrador [BS23], following the approach used in Greyhound. We expect both directions to be interesting future research directions.

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A Lemmas for Discrete Gaussian Distributions

Lemma 4 (Lemma 5 [KLSS23]). For a positive-definite matrix Σ , $\sqrt{\Sigma} \geq \eta_\varepsilon(\Lambda)$ holds if $\|\Sigma^{-1}\|_2 \leq \eta_\varepsilon(\Lambda)^{-2}$.

Lemma 5 (Fact 2.1 [Pei10]). Let $\Sigma_1, \Sigma_2 > 0$ be positive-definite matrices. Define $\Sigma_0^{-1} = \Sigma_1^{-1} + \Sigma_2^{-1} > 0$ and $\Sigma_3 = \Sigma_1 + \Sigma_2 > 0$. Let $\mathbf{x}, \mathbf{c}_1, \mathbf{c}_2 \in \mathbb{R}^n$ be arbitrary, and let $\mathbf{c}_0 \in \mathbb{R}^n$ be such that $\Sigma_0^{-1}\mathbf{c}_0 = \Sigma_1^{-1}\mathbf{c}_1 + \Sigma_2^{-1}\mathbf{c}_2$. Then,

$$\rho_{\sqrt{\Sigma_1}}(\mathbf{x} - \mathbf{c}_1) \cdot \rho_{\sqrt{\Sigma_2}}(\mathbf{x} - \mathbf{c}_2) = \rho_{\sqrt{\Sigma_0}}(\mathbf{x} - \mathbf{c}_0) \cdot \rho_{\sqrt{\Sigma_3}}(\mathbf{c}_1 - \mathbf{c}_2)$$

Lemma 6 (Lemma 10 [HSS24]). Let $k > 1$ be an integer, $\Sigma \in \mathbb{R}^{n \times n}$ be positive-definite matrices, and $\Lambda \subseteq \mathbb{Z}^n$ be an n -dimensional lattice. If $\sqrt{\Sigma/2} \geq \eta_\varepsilon(\Lambda)$ for some $0 < \varepsilon < 1/2$, then for any $\mathbf{u}_0, \dots, \mathbf{u}_{k-1} \in \mathbb{R}^n$, the distribution of $\sum_{k=0}^{t-1} \mathbf{x}_i$, where $\mathbf{x}_i \stackrel{\$}{\leftarrow} \mathcal{D}_{\mathbf{u}_i + \Lambda, \sqrt{\Sigma}}$, is within a statistical distance of $8(k-1)\varepsilon$ from $\mathcal{D}_{\mathbf{u} + \Lambda, \sqrt{k\Sigma}}$, where $\mathbf{u} = \sum_{k=0}^{t-1} \mathbf{u}_i$.

Lemma 7 (Lemma 2.4 [Pei10]). Let $\Lambda \subseteq \mathbb{R}^n$ be any n -dimensional lattice. For any $0 < \varepsilon < 1$, $\Sigma > 0$ such that $\sqrt{\Sigma} \geq \eta_\varepsilon(\Lambda)$, and any $\mathbf{c} \in \mathbb{R}^n$, the following holds

$$\rho_{\sqrt{\Sigma}}(\mathbf{c} + \Lambda) \in \left[\frac{1 - \varepsilon}{1 + \varepsilon}, 1 \right] \cdot \rho_{\sqrt{\Sigma}}(\Lambda).$$

Lemma 8 (Lemma 2.4 [Ban95]). Let $\Lambda \subseteq \mathbb{R}^n$ be any n -dimensional lattice. For any $0 < \varepsilon < 1/3$, $\sigma \geq \eta_\varepsilon(\Lambda)$, and any $\mathbf{c} \in \mathbb{R}^n$,

$$\Pr \left[\|\mathbf{x}\|_\infty > 5\sigma \mid \mathbf{x} \stackrel{\$}{\leftarrow} \mathcal{D}_{\mathbf{c} + \Lambda, \sigma} \right] \leq n \cdot 2^{-111}$$

Lemma 9. Let $\Sigma_0, \Sigma_1, \Sigma_2, \Sigma_3 \in \mathbb{R}^{n \times n}$ be positive-definite matrices, $\mathbf{C}_0, \mathbf{C}_1 \in \mathbb{Z}^{n \times n}$ be invertible matrices, and $\Lambda \subseteq \mathbb{Z}^n$ be an n -dimensional lattice. Suppose

$$\sqrt{(\Sigma_0^{-1} + \mathbf{C}_0^\top \Sigma_1^{-1} \mathbf{C}_0 + \mathbf{C}_1^\top \Sigma_2^{-1} \mathbf{C}_1 + (\mathbf{C}_0 \mathbf{C}_1)^\top \Sigma_3^{-1} \mathbf{C}_0 \mathbf{C}_1)^{-1}} \geq \eta_\varepsilon(\Lambda)$$

for some $\varepsilon > 0$. Then, the following two distributions are within statistical distance ε for all $\mathbf{u} \in \mathbb{Z}^n$.

$$\left\{ \begin{array}{l} (\mathbf{y}_0, \mathbf{y}_1, \mathbf{y}_2) \\ \left| \begin{array}{l} \mathbf{x}_0 \stackrel{\$}{\leftarrow} \mathcal{D}_{\Lambda, \sqrt{\Sigma_0}}, \mathbf{x}_1 \stackrel{\$}{\leftarrow} \mathcal{D}_{\Lambda, \sqrt{\Sigma_1}} \\ \mathbf{x}_2 \stackrel{\$}{\leftarrow} \mathcal{D}_{\Lambda, \sqrt{\Sigma_2}}, \mathbf{x}_3 \stackrel{\$}{\leftarrow} \mathcal{D}_{\Lambda, \sqrt{\Sigma_3}}, \\ \mathbf{y}_0 = \mathbf{x}_1 + \mathbf{C}_0 \mathbf{x}_0, \mathbf{y}_1 = \mathbf{x}_2 + \mathbf{C}_1 \mathbf{x}_0, \mathbf{y}_2 = \mathbf{x}_3 + \mathbf{C}_1 \mathbf{x}_1 \end{array} \right. \end{array} \right\}$$

$$\left\{ \begin{array}{l} (\mathbf{y}'_0, \mathbf{y}'_1, \mathbf{y}'_2) \\ \left| \begin{array}{l} \mathbf{x}'_0 \stackrel{\$}{\leftarrow} \mathcal{D}_{\mathbf{u} + \Lambda, \sqrt{\Sigma_0}}, \mathbf{x}'_1 \stackrel{\$}{\leftarrow} \mathcal{D}_{-\mathbf{C}_0 \mathbf{u} + \Lambda, \sqrt{\Sigma_1}} \\ \mathbf{x}'_2 \stackrel{\$}{\leftarrow} \mathcal{D}_{-\mathbf{C}_1 \mathbf{u} + \Lambda, \sqrt{\Sigma_2}}, \mathbf{x}'_3 \stackrel{\$}{\leftarrow} \mathcal{D}_{\mathbf{C}_1 \mathbf{C}_0 \mathbf{u} + \Lambda, \sqrt{\Sigma_3}}, \\ \mathbf{y}'_0 = \mathbf{x}'_1 + \mathbf{C}_0 \mathbf{x}'_0, \mathbf{y}'_1 = \mathbf{x}'_2 + \mathbf{C}_1 \mathbf{x}'_0, \mathbf{y}'_2 = \mathbf{x}'_3 + \mathbf{C}_1 \mathbf{x}'_1 \end{array} \right. \end{array} \right\}$$

Proof. We first note that the sample spaces of the two distributions above are both equal to $\Lambda \times \Lambda \times \Lambda$. Thus, it suffices to prove the following holds for all $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2 \in \Lambda$.

$$\begin{aligned} & \Pr[\mathbf{y}'_0 = \mathbf{w}_0, \mathbf{y}'_1 = \mathbf{w}_1, \mathbf{y}'_2 = \mathbf{w}_2] \\ & \in \left[\frac{1-\varepsilon}{1+\varepsilon}, 1 \right] \cdot \Pr[\mathbf{y}_0 = \mathbf{w}_0, \mathbf{y}_1 = \mathbf{w}_1, \mathbf{y}_2 = \mathbf{w}_2] \end{aligned}$$

By definition, the following hold for $\mathbf{S}_1 = \mathbf{C}_0^{-1} \boldsymbol{\Sigma}_1 (\mathbf{C}_0^\top)^{-1}$, $\mathbf{S}_2 = \mathbf{C}_1^{-1} \boldsymbol{\Sigma}_2 (\mathbf{C}_1^\top)^{-1}$, and $\mathbf{S}_3 = (\mathbf{C}_1 \mathbf{C}_0)^{-1} \boldsymbol{\Sigma}_3 (\mathbf{C}_0^\top \mathbf{C}_1^\top)^{-1}$.

$$\begin{aligned} & \Pr[\mathbf{y}'_0 = \mathbf{w}_0, \mathbf{y}'_1 = \mathbf{w}_1, \mathbf{y}'_2 = \mathbf{w}_2] \\ & = \sum_{\mathbf{v} \in \mathbf{u} + \Lambda} \left(\Pr[\mathbf{x}'_0 = \mathbf{v}] \cdot \Pr[\mathbf{x}'_1 = \mathbf{w}_0 - \mathbf{C}_0 \mathbf{v}] \right. \\ & \quad \left. \cdot \Pr[\mathbf{x}'_2 = \mathbf{w}_1 - \mathbf{C}_1 \mathbf{v}] \cdot \Pr[\mathbf{x}'_3 = \mathbf{w}_2 - \mathbf{C}_1 \mathbf{w}_0 + \mathbf{C}_1 \mathbf{C}_0 \mathbf{v}] \right) \\ & \propto \sum_{\mathbf{v} \in \mathbf{u} + \Lambda} \left(\rho_{\sqrt{\boldsymbol{\Sigma}_0}}(\mathbf{v}) \cdot \rho_{\sqrt{\boldsymbol{\Sigma}_1}}(\mathbf{w}_0 - \mathbf{C}_0 \mathbf{v}) \right. \\ & \quad \left. \cdot \rho_{\sqrt{\boldsymbol{\Sigma}_2}}(\mathbf{w}_1 - \mathbf{C}_1 \mathbf{v}) \cdot \rho_{\sqrt{\boldsymbol{\Sigma}_3}}(\mathbf{w}_2 - \mathbf{C}_1 \mathbf{w}_0 + \mathbf{C}_1 \mathbf{C}_0 \mathbf{v}) \right) \\ & = \sum_{\mathbf{v} \in \mathbf{u} + \Lambda} \left(\rho_{\sqrt{\boldsymbol{\Sigma}_0}}(\mathbf{v}) \cdot \rho_{\sqrt{\mathbf{S}_1}}(\mathbf{v} - \mathbf{C}_0^{-1} \mathbf{w}_0) \right. \\ & \quad \left. \cdot \rho_{\sqrt{\mathbf{S}_2}}(\mathbf{v} - \mathbf{C}_1^{-1} \mathbf{w}_1) \cdot \rho_{\sqrt{\mathbf{S}_3}}(\mathbf{v} - \mathbf{C}_0^{-1} \mathbf{w}_0 + \mathbf{C}_0^{-1} \mathbf{C}_1^{-1} \mathbf{w}_1) \right) \end{aligned}$$

By applying Lemma 5 sequentially, the above can be written as follows for some $\mathbf{a}_0, \mathbf{a}_1, \mathbf{b}_0, \dots, \mathbf{b}_3 \in \mathbb{R}^n$ that are independent of $\mathbf{v}$

$$\begin{aligned} & = \rho_{\sqrt{\mathbf{U}_0}}(\mathbf{b}_0) \cdot \left(\sum_{\mathbf{v} \in \mathbf{u} + \Lambda} \rho_{\sqrt{(\boldsymbol{\Sigma}_0^{-1} + \mathbf{S}_1^{-1})^{-1}}}(\mathbf{v} - \mathbf{a}_0) \cdot \rho_{\sqrt{\mathbf{S}_2}}(\mathbf{v} - \mathbf{C}_1^{-1} \mathbf{w}_1) \right. \\ & \quad \left. \cdot \rho_{\sqrt{\mathbf{S}_3}}(\mathbf{v} - \mathbf{C}_0^{-1} \mathbf{w}_0 + \mathbf{C}_0^{-1} \mathbf{C}_1^{-1} \mathbf{w}_1) \right) \\ & = \rho_{\sqrt{\mathbf{U}_0}}(\mathbf{b}_0) \cdot \rho_{\sqrt{\mathbf{U}_1}}(\mathbf{b}_1) \cdot \left(\sum_{\mathbf{v} \in \mathbf{u} + \Lambda} \rho_{\sqrt{(\boldsymbol{\Sigma}_0^{-1} + \mathbf{S}_1^{-1} + \mathbf{S}_2^{-1})^{-1}}}(\mathbf{v} - \mathbf{a}_1) \right. \\ & \quad \left. \cdot \rho_{\sqrt{\mathbf{S}_3}}(\mathbf{v} - \mathbf{C}_0^{-1} \mathbf{w}_0 + \mathbf{C}_0^{-1} \mathbf{C}_1^{-1} \mathbf{w}_1) \right) \\ & = \rho_{\sqrt{\mathbf{U}_0}}(\mathbf{b}_0) \cdot \rho_{\sqrt{\mathbf{U}_1}}(\mathbf{b}_1) \cdot \rho_{\sqrt{\mathbf{U}_2}}(\mathbf{b}_2) \cdot \sum_{\mathbf{v} \in \mathbf{u} + \Lambda} \rho_{\sqrt{\mathbf{U}_3}}(\mathbf{v} - \mathbf{b}_3) \end{aligned}$$

where

$$\begin{aligned} \mathbf{U}_0 & = \boldsymbol{\Sigma}_0 + \mathbf{S}_1 \\ \mathbf{U}_1 & = (\boldsymbol{\Sigma}_0^{-1} + \mathbf{S}_1^{-1})^{-1} + \mathbf{S}_2 \\ \mathbf{U}_2 & = (\boldsymbol{\Sigma}_0^{-1} + \mathbf{S}_1^{-1} + \mathbf{S}_2^{-1})^{-1} + \mathbf{S}_3, \\ \mathbf{U}_3 & = (\boldsymbol{\Sigma}_0^{-1} + \mathbf{S}_1^{-1} + \mathbf{S}_2^{-1} + \mathbf{S}_3^{-1})^{-1} \end{aligned}$$

In the same way, the following also holds.

$$\begin{aligned} & \Pr[\mathbf{y}_0 = \mathbf{w}_0, \mathbf{y}_1 = \mathbf{w}_1, \mathbf{y}_2 = \mathbf{w}_2] \\ & \propto \rho_{\sqrt{\mathbf{U}_0}}(\mathbf{b}_0) \cdot \rho_{\sqrt{\mathbf{U}_1}}(\mathbf{b}_1) \cdot \rho_{\sqrt{\mathbf{U}_2}}(\mathbf{b}_2) \cdot \sum_{\mathbf{v} \in \Lambda} \rho_{\sqrt{\mathbf{U}_3}}(\mathbf{v} - \mathbf{b}_3) \end{aligned}$$

Since we assume $\sqrt{\mathbf{U}_3} \geq \eta_\varepsilon(\Lambda)$, the following holds due to Lemma 7.

$$\sum_{\mathbf{v} \in \mathbf{u} + \Lambda} \rho_{\sqrt{\mathbf{U}_3}}(\mathbf{v} - \mathbf{b}_3) \in \left[\frac{1 - \varepsilon}{1 + \varepsilon}, 1 \right] \cdot \sum_{\mathbf{v} \in \Lambda} \rho_{\sqrt{\mathbf{U}_3}}(\mathbf{v} - \mathbf{b}_3)$$

Therefore, we can conclude the following

$$\begin{aligned} & \Pr[\mathbf{y}'_0 = \mathbf{w}_0, \mathbf{y}'_1 = \mathbf{w}_1, \mathbf{y}'_2 = \mathbf{w}_2] \\ & \in \left[\frac{1 - \varepsilon}{1 + \varepsilon}, 1 \right] \cdot \Pr[\mathbf{y}_0 = \mathbf{w}_0, \mathbf{y}_1 = \mathbf{w}_1, \mathbf{y}_2 = \mathbf{w}_2] \end{aligned}$$

which implies that the two distributions are within statistical distance ε . $\square$

B Coordinate-wise Special Soundness

Definition 14 (Coordinate-Wise Special Soundness³). Let $\Pi = (\text{Setup}, \mathbf{P}, \mathbf{V})$ be a public-coin three-round interactive proof system for relation $\mathbf{R}$, where the verifier chooses uniformly random challenges from the set S^ℓ , and let $(a, \mathbf{c}^{(k)}, z^{(k)})_{0 \leq k \leq \ell}$ be $\ell + 1$ accepting transcripts, with a common first message a , responses $z^{(k)}$, and challenges $\mathbf{c}^{(k)}$ such that $\mathbf{c}^{(j)}$ differs from $\mathbf{c}^{(n)}$ only at the j -th coordinate for $0 \leq j < \ell$. We say that Π is ℓ -coordinate-wise special sound if there exists a polynomial-time algorithm that, on input public parameters $\mathbf{pp}$ and $(a, \mathbf{c}^{(k)}, z^{(k)})_{0 \leq k \leq \ell}$, outputs a witness $\mathbf{w}$ such that $(\mathbf{x}, \mathbf{w}) \in \mathbf{R}$.

Lemma 10 (Lemma 2.31 [FMN24]). Let Π be a public-coin three-round protocol. If Π is ℓ -coordinate-wise special sound for a challenge set S^ℓ , then it is knowledge sound if $\ell/|S| = \text{negl}(\lambda)$.

C Deferred Proofs

C.1 Proof of Theorem 2

Proof. Let $\Pi_1 = (\mathbf{P}_1, \mathbf{V}_1)$ and $\Pi_2 = (\mathbf{P}_2, \mathbf{V}_2)$, and let $\Pi_2 \circ \Pi_1 = (\mathbf{P} := \mathbf{P}_2 \circ \mathbf{P}_1, \mathbf{V} := \mathbf{V}_2 \circ \mathbf{V}_1)$. Below, we prove the relaxed completeness and knowledge soundness of $(\mathbf{P}, \mathbf{V})$.

Relaxed Completeness. Let $(\mathbf{x}_1, \mathbf{w}_1) \in \mathbf{R}_1$, and let $(\mathbf{x}_2, \mathbf{w}_2)$ be an output of $\mathbf{P}_1$ after an interaction with $\mathbf{V}_1$ for the input $(\mathbf{x}_1, \mathbf{w}_1)$. By the relaxed completeness of Π_1 , it holds that $(\mathbf{x}_2, \mathbf{w}_2) \in \mathbf{R}_2$ with overwhelming probability. Next,

³ A more general version can be found in [FMN24].

by the relaxed completeness of Π_2 , the verifier V_2 outputs 1 with overwhelming probability. Therefore, the composed verifier V outputs 1 with overwhelming probability when $(x_1, w_1) \in R_1$. Thus, the relaxed completeness property holds for (P, V) .

Knowledge Soundness. Let A and P^* be PPT adversaries for the knowledge soundness of the composed protocol (P, V) , which generate accepting transcripts with non-negligible probability. To prove knowledge soundness, we construct an expected PPT extractor E that outputs an instance-witness pair $(x_1, w_1) \in R'_1$ with overwhelming probability.

Let $(x_1, st_1) \leftarrow A(pp)$. The interaction between P^* and V can be described as follows for some PPT algorithms P_1^* and P_2^* , since V is the composition of V_1 and V_2 .

$$P^*(pp, st_1) \rightarrow \perp$$

1. $(tr_1, st_2, x_2) \leftarrow \langle P_1^*(pp, st_1), V_1(pp, x_1) \rangle$.
2. $(tr_2, \perp, b) \leftarrow \langle P_2^*(pp, st_2), V_2(pp, x_2) \rangle$

Then we construct a PPT adversary A_2 as follows, so that A_2 and P_2^* act as adversaries against the knowledge soundness of Π_2 .

$$A_2^*(pp) \rightarrow (st_2, x_2)$$

1. $(st_1, x_1) \leftarrow A(pp)$.
2. $(tr_1, st_2, x_2) \leftarrow \langle P_1^*(pp, st_1), V_1(pp, x_1) \rangle$.
3. Output (st_2, x_2) .

Since P^* generates accepting transcripts with non-negligible probability, P_2^* also generates accepting transcripts with non-negligible probability. Thus, by the knowledge soundness of Π_2 , there exists an expected PPT extractor E_2 that outputs a valid witness w_2 with overwhelming probability when P_2^* generates accepting transcripts. We then construct an adversary P_1^{**} as follows, so that P_1^{**} and A act as adversaries against the knowledge soundness of Π_1 .

$$P_1^{**}(pp, st_1) \rightarrow (x_2, w_2)$$

1. $(tr_1, st_2, x_2) \leftarrow \langle P_1^*(pp, st_1), V_1(pp, x_1) \rangle$.
2. $x_2 \leftarrow \phi_1(pp, tr_1, x_1)$
3. $w_2 \leftarrow E_2(pp, st_2, x_2)$.
4. Output (x_2, w_2) .

We note that the output $(x_2, w_2) \in R'_2$ occurs with non-negligible probability due to the assumption on P_1^* . Thus, by the knowledge soundness of Π_1 , there exists an extractor E_1 for A and P_1^* that outputs a valid witness w_1 with

overwhelming probability. We remark that E_1 is also an extractor for A and P^* . Therefore, the protocol (P, V) satisfies knowledge soundness. $\square$

C.2 Proof of Theorem 3

Proof. Suppose there exists an adversary A against the binding property with non-negligible advantage. Then, it outputs $(\hat{u}, f, f', \text{op}, \text{op}')$ such that $f \neq f'$, $\text{Open}(\text{pp}, \hat{u}, f, \text{op}) = 1$, and $\text{Open}(\text{pp}, \hat{u}, f', \text{op}') = 1$.

Let $\text{op} = (\hat{\mathbf{F}}, \mathbf{R}, \hat{\mathbf{t}}, \mathbf{c}, \alpha)$, $\mathbf{s} := \mathbf{D}\hat{\mathbf{t}} - q_s\alpha \cdot \hat{\mathbf{u}}$, $\text{op}' = (\hat{\mathbf{F}}', \mathbf{R}', \hat{\mathbf{t}}', \mathbf{c}', \alpha')$, and $\mathbf{s}' := \mathbf{D}\hat{\mathbf{t}}' - q_s\alpha' \cdot \hat{\mathbf{u}}$. Then, we have the following.

$$\begin{aligned} [\mathbf{D} \mid \mathbf{I}_\kappa] \begin{bmatrix} \alpha'\hat{\mathbf{t}} - \alpha\hat{\mathbf{t}}' \\ \alpha\mathbf{s}' - \alpha'\mathbf{s} \end{bmatrix} &= \mathbf{0} \pmod{q_o} \\ \|\alpha'\hat{\mathbf{t}} - \alpha\hat{\mathbf{t}}'\|_2 + \|\alpha\mathbf{s}' - \alpha'\mathbf{s}\|_2 &\leq 2B_\alpha B_o \end{aligned}$$

Thus, we have $\alpha'\hat{\mathbf{t}} = \alpha\hat{\mathbf{t}}'$ due to the hardness of $\text{MSIS}_{q_o, \kappa, 2B_\alpha B_o}$.

Next, let $\mathbf{h}_j := \mathbf{A}\hat{\mathbf{f}}_j + \mathbf{B}\mathbf{r}_j - q_h c_j \cdot \hat{\mathbf{t}}_j$ and $\mathbf{h}'_j := \mathbf{A}\hat{\mathbf{f}}'_j + \mathbf{B}\mathbf{r}'_j - q_h c'_j \cdot \hat{\mathbf{t}}'_j$ for $0 \leq j \leq n$. Then, we have the following for $0 \leq j < n$.

$$\begin{aligned} \alpha'c'_j\hat{\mathbf{h}}_j - \alpha c_j\hat{\mathbf{h}}'_j &= \mathbf{A}(\alpha'c'_j\hat{\mathbf{f}}_j - \alpha c_j\hat{\mathbf{f}}'_j) + \mathbf{B}(\alpha'c'_j\mathbf{r}_j - \alpha c_j\mathbf{r}'_j) \\ \implies [\mathbf{A} \mid \mathbf{B} \mid \mathbf{I}_\mu] \left(\begin{bmatrix} \alpha'c'_j\hat{\mathbf{f}}_j - \alpha c_j\hat{\mathbf{f}}'_j \\ \alpha'c'_j\mathbf{r}_j - \alpha c_j\mathbf{r}'_j \end{bmatrix} - \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \alpha'c'_j\hat{\mathbf{h}}_j - \alpha c_j\hat{\mathbf{h}}'_j \end{bmatrix} \right) &= \mathbf{0} \pmod{q} \end{aligned}$$

Also, the following norm bound holds for $0 \leq j < n$.

$$\|\alpha'c'_j\hat{\mathbf{f}}_j - \alpha c_j\hat{\mathbf{f}}'_j\|_2 + \|\alpha'c'_j\mathbf{r}_j - \alpha c_j\mathbf{r}'_j\|_2 + \|\alpha'c'_j\hat{\mathbf{h}}_j - \alpha c_j\hat{\mathbf{h}}'_j\|_2 \leq 2B_\alpha B_c B$$

Thus, by the hardness of $\text{MSIS}_{q, \mu, 2B_\alpha B_c B}$, we have $\alpha'c'_j\hat{\mathbf{f}}_j = \alpha c_j\hat{\mathbf{f}}'_j$ for all $0 \leq j < n$. This implies

$$\begin{aligned} f_{i,j} &= \text{Dcd}(\alpha c_j)^{-1} \cdot \text{Dcd}(\hat{f}_{i,j}) \\ &= \text{Dcd}(\alpha'c'_j)^{-1} \cdot \text{Dcd}(\hat{f}'_{i,j}) = f'_{i,j}. \end{aligned}$$

Therefore, $f = f'$, which contradicts the assumption that $f \neq f'$. Hence, no such PPT adversary A exists for the binding property. $\square$

C.3 Proof of Theorem 4

Proof. To prove the hiding property, it suffices to consider only the distribution of $\mathbf{t}_j = \mathbf{A}\hat{\mathbf{f}}_j + \mathbf{B}\mathbf{r}_j$ for $0 \leq j \leq n$. We note that $\mathbf{B}\mathbf{r}_j$ is a MLWE sample from $\text{MLWE}_{q, \nu, \tau_0}$ and $\mathbf{B}\mathbf{r}_n$ is a MLWE sample from $\text{MLWE}_{q, \nu, \sqrt{n+1} \cdot \tau_1}$. Thus, if both $\text{MLWE}_{q, \nu, \tau_0}$ and $\text{MLWE}_{q, \nu, \sqrt{n+1} \cdot \tau_1}$ are computationally hard, then $\mathbf{t}_j$ becomes computationally indistinguishable from a random sample from R_q^ν for $0 \leq j \leq n$. $\square$

C.4 Proof of Theorem 5

Proof. Below, we prove relaxed completeness and knowledge soundness of (Eval.P, Eval.V).

Relaxed Completeness. It directly follows from the observation since $\bar{\mathbf{f}} = \sum_{j=0}^{n-1} c_j \hat{\mathbf{f}}_j + \hat{\mathbf{f}}_n$, $\bar{\mathbf{r}} = \sum_{j=0}^{n-1} c_j \mathbf{r}_j + \mathbf{r}_n$, and $\bar{\mathbf{h}} = \sum_{j=0}^{n-1} c_j \mathbf{h}_j + \mathbf{h}_n$ hold.

Knowledge soundness. By Lemma 10, it suffices to prove coordinate-wise special soundness. Let $(\hat{\mathbf{t}}, \mathbf{z}, \mathbf{c}^{(j)}, \bar{\mathbf{f}}^{(j)}, \bar{\mathbf{r}}^{(j)})$ be $n+1$ accepting transcripts for an instance $\mathbf{x} = (\hat{\mathbf{u}}, x, y)$, where $\mathbf{c}^{(j)}$ differs from $\mathbf{c}^{(n)}$ only at the j -th coordinate for $0 \leq j < n$. Let $\mathbf{s} := \mathbf{D}\hat{\mathbf{t}} - q_s \hat{\mathbf{u}}$. Then the following holds by definition.

$$\|\mathbf{s}\|_2 + \|\hat{\mathbf{t}}\|_2 \leq B_o = B'_o$$

Next, let $\bar{\mathbf{h}}^{(j)} := \mathbf{A}\bar{\mathbf{f}}^{(j)} + \mathbf{B}\bar{\mathbf{r}}^{(j)} - q_h \cdot \hat{\mathbf{T}} \begin{bmatrix} \mathbf{c}^{(j)} \\ 1 \end{bmatrix}$ for $0 \leq j \leq n$, and define c_j , $\bar{\mathbf{f}}_j$, $\bar{\mathbf{r}}_j$, and $\bar{\mathbf{h}}_j$ as follows for $0 \leq j < n$.

$$\begin{aligned} c_j &:= c_j^{(j)} - c_j^{(n)}, & \bar{\mathbf{f}}_j &:= \bar{\mathbf{f}}^{(j)} - \bar{\mathbf{f}}^{(n)} \\ \bar{\mathbf{r}}_j &:= \bar{\mathbf{r}}^{(j)} - \bar{\mathbf{r}}^{(n)}, & \bar{\mathbf{h}}_j &:= \bar{\mathbf{h}}^{(j)} - \bar{\mathbf{h}}^{(n)} \end{aligned}$$

Then, the following holds for $0 \leq j < n$.

$$\begin{aligned} \bar{\mathbf{h}}_j &:= \mathbf{A}\bar{\mathbf{f}}_j + \mathbf{B}\bar{\mathbf{r}}_j - q_h c_j \cdot \hat{\mathbf{t}}_j \\ \|c_j\|_1 &\leq 2B_C, \quad \|\bar{\mathbf{f}}_j\|_2 + \|\bar{\mathbf{r}}_j\|_2 + \|\bar{\mathbf{h}}_j\|_2 \leq 2\bar{B} \end{aligned}$$

For $\bar{\mathbf{f}}_j = (\bar{f}_{0,j}, \dots, \bar{f}_{m,j}) \in R^{m+1}$, define $f(X) \in \mathbb{Z}_p^{<N}[X]$ as follows, where $f_{i,j} := \text{Dcd}(c_j)^{-1} \cdot \text{Dcd}(\bar{f}_{i,j})$.

$$f(X) := \begin{bmatrix} 1 & x^n & \dots & x^{N-n} & x \end{bmatrix} \begin{bmatrix} f_{0,0} & \dots & f_{0,n-1} \\ \vdots & & \vdots \\ f_{m,0} & \dots & f_{m,n-1} \end{bmatrix} \begin{bmatrix} 1 \\ x \\ \vdots \\ x^{n-1} \end{bmatrix}$$

Then, it holds that $\text{Open}(\text{pp}, f, \hat{\mathbf{u}}, \text{op}) = 1$ for $\text{op} = (\bar{\mathbf{F}}, \bar{\mathbf{R}}, \hat{\mathbf{t}}, \mathbf{c}, 1)$, where $\bar{\mathbf{F}} = [\bar{\mathbf{f}}_0 \mid \dots \mid \bar{\mathbf{f}}_{n-1} \mid \mathbf{0}]$, $\bar{\mathbf{R}} = [\bar{\mathbf{r}}_0 \mid \dots \mid \bar{\mathbf{r}}_{n-1} \mid \mathbf{0}]$, and $\mathbf{c} = (c_0, \dots, c_{n-1})$.

Lastly, it remains to show that $f(x) = y$. Let $\mathbf{z} = (z_0, \dots, z_n)$, then the following holds for $0 \leq j < n$.

$$\hat{\mathbf{x}}^\top \bar{\mathbf{f}}_j = \mathbf{z}^\top \begin{bmatrix} \mathbf{c}^{(j)} - \mathbf{c}^{(n)} \\ 0 \end{bmatrix} = c_j \cdot z_j$$

Then, the following holds for $c := \prod_{j=0}^{n-1} c_j$.

$$\hat{\mathbf{x}}^\top \left[(c/c_j) \cdot \bar{\mathbf{f}}_0 \mid \dots \mid (c/c_{n-1}) \cdot \bar{\mathbf{f}}_{n-1} \mid \mathbf{0} \right] \hat{\mathbf{x}} = c \cdot \mathbf{z}^\top \hat{\mathbf{x}}$$

We note that the following holds for each $f_{i,j}$.

$$f_{i,j} = \text{Dcd}(c_j)^{-1} \cdot \text{Dcd}(\bar{f}_{i,j}) = \text{Dcd}(c)^{-1} \cdot \text{Dcd}(c/c_j \cdot \bar{f}_{i,j}).$$

Thus, we have the following.

$$\begin{aligned} f(x) &= \text{Dcd}(c)^{-1} \cdot \text{Dcd} \left(\hat{\mathbf{x}}^\top \left[(c/c_0) \cdot \bar{\mathbf{f}}_0 \mid \cdots \mid (c/c_{n-1}) \cdot \bar{\mathbf{f}}_{n-1} \mid \mathbf{0} \right] \hat{\mathbf{x}} \right) \\ &= \text{Dcd}(c)^{-1} \cdot \text{Dcd}(c \cdot \mathbf{z}^\top \hat{\mathbf{x}}) = y \end{aligned}$$

Therefore, $\left((\hat{\mathbf{u}}, x, y), (f, \text{op}) \right) \in \text{R}_{\text{PCS}}^{(B'_j, B'_o, B'_c)}$ holds for $B' = 2\bar{B}$, $B'_o = B_o$, and $B'_c = 2B_c$. $\square$

C.5 Proof of Theorem 6

Proof. By Lemma 2, we have the following norm bounds for each $\hat{f}_{i,j}$.

- $\|\hat{f}_{0,j}\|_\infty \leq 5(b+1) \cdot \sigma_0 \sqrt{m+1}$ for $0 \leq j < n$.
- $\|\hat{f}_{i,j}\|_\infty \leq 5(b+1) \cdot \sigma_1$ for $1 \leq i \leq m$ and $0 \leq j < n$.
- $\|\hat{f}_{0,n}\|_\infty \leq 5(b+1) \cdot \sigma_2 \sqrt{mn+1}$.
- $\|\hat{f}_{i,n}\|_\infty \leq 5(b+1) \cdot \sigma_3 \sqrt{n+1}$ for $1 \leq i \leq m$.

Then, the following holds for each $\hat{\mathbf{f}}_j$.

- $\|\hat{\mathbf{f}}_j\|_2 \leq 5(b+1) \sqrt{(\sigma_0^2 + \sigma_1^2)(m+1)d}$ for $0 \leq j < n$.
- $\|\hat{\mathbf{f}}_n\|_2 \leq 5(b+1) \sqrt{(\sigma_2^2 + \sigma_3^2)(m+1)(n+1)d}$.

By Lemma 8, we have the following norm bounds for each $\mathbf{r}_j$.

- $\|\mathbf{r}_j\|_2 \leq \tau_0 \sqrt{(\mu + \nu)d}$ for $0 \leq j < n$.
- $\|\mathbf{r}_n\|_2 \leq \tau_1 \sqrt{(n+1)(\mu + \nu)d}$.

Finally, we have the following norm bounds for each $\mathbf{h}_j$, $\hat{\mathbf{t}}_j$, and $\mathbf{s}_j$ by definition.

- $\|\mathbf{h}_j\|_2 \leq q_h \sqrt{\mu d}$
- $\|\hat{\mathbf{t}}_j\|_2 \leq (q/q_h) \sqrt{\mu n d}$
- $\|\mathbf{s}_j\|_2 \leq q_s \sqrt{\kappa d}$

Therefore, the following norm bounds hold.

- $\|\hat{\mathbf{f}}_j\|_2 + \|\mathbf{r}_j\|_2 + \|\mathbf{h}_j\|_2 \leq B_j$ for $0 \leq j < n$.
- $\|\hat{\mathbf{f}}_n\|_2 + \|\mathbf{r}_n\|_2 + \|\mathbf{h}_n\|_2 \leq B_n$.
- $\|\mathbf{s}_j\|_2 + \|\hat{\mathbf{t}}_j\|_2 \leq B_o$ for $0 \leq j \leq n$.

$\square$

C.6 Proof of Theorem 7

Proof. To prove the hiding property, we simulate the distribution of $(\hat{\mathbf{u}}, \hat{\mathbf{t}}, \mathbf{z}, \mathbf{c}, \bar{\mathbf{f}}, \bar{\mathbf{r}})$ for all $f \in \mathbb{Z}_p^{<N}[X]$ and $x \in \mathbb{Z}_p$. We prove the hiding property using hybrid arguments as follows.

Hybrid 1: We define the first hybrid distribution as follows.

$$\mathcal{H}_1(\text{pp}, f, x, y) \rightarrow (\hat{\mathbf{u}}, \hat{\mathbf{t}}, \mathbf{z}, \mathbf{c}, \bar{\mathbf{f}}, \bar{\mathbf{r}})$$

1. $(_, \text{op}) \leftarrow \text{Com}(\text{pp}, f)$, where $\text{op} = (\hat{\mathbf{F}}, \mathbf{R}, \hat{\mathbf{t}}, \mathbf{1}, 1)$.
2. $\mathbf{c} = (c_0, \dots, c_{n-1}) \xleftarrow{\$} \text{Ecd}(\mathcal{C})^n$.
3. $\hat{f}_{0,n}^{(i,j)} \xleftarrow{\$} \text{REcd}_{\sigma_2}(\text{Dcd}(\hat{x}_i c_j \cdot \hat{f}_{i,j}))$
for $1 \leq i \leq m$, $0 \leq j < n$, and $\hat{\mathbf{x}} = (\hat{x}_0, \dots, \hat{x}_m)$.
4. $\hat{f}_{0,n}^* \xleftarrow{\$} \text{REcd}_{\sigma_2}(f_{0,n})$ for $f_{0,n} \xleftarrow{\$} \mathbb{Z}_p$.
5. $\hat{f}_{0,n} := \hat{f}_{0,n}^* + \sum_{i=1}^m \sum_{j=0}^{n-1} \hat{f}_{0,n}^{(i,j)}$.
6. $\hat{f}_{i,n}^{(j)} \xleftarrow{\$} \text{REcd}_{\sigma_3}(-\text{Dcd}(c_j \cdot \hat{f}_{i,j}))$ for $1 \leq i \leq m$ and $0 \leq j < n$.
7. $\hat{f}_{i,n}^* \xleftarrow{\$} \text{REcd}_{\sigma_3}(f_{i,n})$ for $f_{i,n} \xleftarrow{\$} \mathbb{Z}_p$ and $1 \leq i \leq m$.
8. $\hat{f}_{i,n} := \hat{f}_{i,n}^* + \sum_{j=0}^{n-1} \hat{f}_{i,n}^{(j)}$ for $1 \leq i \leq m$.
9. Redefine $\hat{\mathbf{f}}_n := (\hat{f}_{0,n}, \dots, \hat{f}_{m,n})$.
10. $\mathbf{r}_n^{(0)}, \dots, \mathbf{r}_n^{(n-1)}, \mathbf{r}_n^* \xleftarrow{\$} \mathcal{D}_{\mathbb{Z}^d, \tau_1}^{\mu+\nu}$.
11. Redefine $\mathbf{r}_n := \mathbf{r}_n^* + \sum_{j=0}^{n-1} \mathbf{r}_n^{(j)}$.
12. Redefine $\hat{\mathbf{t}}_n := \lfloor (\mathbf{A}\hat{\mathbf{f}}_n + \mathbf{B}\mathbf{r}_n)/q_h \rfloor$.
13. $\mathbf{z}^\top := \hat{\mathbf{x}}^\top \hat{\mathbf{F}}$, $\bar{\mathbf{f}} := \hat{\mathbf{F}} \begin{bmatrix} \mathbf{c} \\ 1 \end{bmatrix}$, $\mathbf{r} := \mathbf{R} \begin{bmatrix} \mathbf{c} \\ 1 \end{bmatrix}$, and $\hat{\mathbf{u}} := \lfloor \mathbf{D}\hat{\mathbf{t}}/q_s \rfloor$.

Fig. 6: Hybrid Distribution 1

In the first hybrid distribution $\mathcal{H}_1$, the difference between the real transcript tr lies in how $\hat{\mathbf{f}}_n$ and $\mathbf{r}_n$ are constructed, as described in the underlined parts in Fig. 6.

For $\hat{f}_{0,n}$, it follows the following distribution in the real transcript tr ,

$$\mathcal{D}_{\text{Ecd}(f_{0,n}) + \mathbf{T}\mathbb{Z}^d, \sqrt{mn+1} \cdot \sigma_2 \mathbf{T}}$$

while it follows the following distribution in the hybrid distribution $\mathcal{H}_1$.

$$\sum_{i=1}^m \sum_{j=0}^{n-1} \mathcal{D}_{\hat{x}_i c_j \cdot \hat{f}_{i,j} + \mathbf{T}\mathbb{Z}^d, \sigma_2 \mathbf{T}} + \mathcal{D}_{\text{Ecd}(f_{0,n}) + \mathbf{T}\mathbb{Z}^d, \sigma_2 \mathbf{T}}$$

By Lemma 6, the latter distribution is statistically close to the following distribution

$$\mathcal{D}_{\sum_{i=1}^m \sum_{j=0}^{n-1} \hat{x}_i c_j \cdot \hat{f}_{i,j} + \text{Ecd}(f_{0,n}) + \mathbf{T}\mathbb{Z}^d, \sqrt{mn+1} \cdot \sigma_2 \mathbf{T}}$$

We note that both $\text{Ecd}(f_{0,n})$ in tr and $\sum_{i=1}^m \sum_{j=0}^{n-1} \hat{x}_i c_j \cdot \hat{f}_{i,j} + \text{Ecd}(f_{0,n})$ in $\mathcal{H}_1$ follow the identical distribution modulo $\mathbf{T}\mathbb{Z}^d$ since $f_{0,n} \xleftarrow{\$} \mathbb{Z}_p$. Thus, $\hat{f}_{0,n}$ in tr and $\mathcal{H}_1$ are statistically close.

For $\hat{f}_{i,n}$, where $1 \leq i \leq m$, it follows the following distribution in the real transcript tr ,

$$\mathcal{D}_{\text{Ecd}(f_{i,n}) + \mathbf{T}\mathbb{Z}^d, \sqrt{n+1} \cdot \sigma_3 \mathbf{T}}$$

while it follows the following distribution in the hybrid distribution $\mathcal{H}_1$.

$$\sum_{j=0}^{n-1} \mathcal{D}_{c_j \cdot \hat{f}_{i,j} + \mathbf{T}\mathbb{Z}^d, \sigma_3 \mathbf{T}} + \mathcal{D}_{\text{Ecd}(f_{i,n}) + \mathbf{T}\mathbb{Z}^d, \sigma_3 \mathbf{T}}$$

By Lemma 6, the latter distribution is statistically close to the following distribution.

$$\mathcal{D}_{\sum_{j=0}^{n-1} c_j \cdot \hat{f}_{i,j} + \text{Ecd}(f_{i,n}) + \mathbf{T}\mathbb{Z}^d, \sqrt{n+1} \cdot \sigma_2 \mathbf{T}}$$

We note that both $\text{Ecd}(f_{i,n})$ in tr and $\sum_{j=0}^{n-1} c_j \cdot \hat{f}_{i,j} + \text{Ecd}(f_{i,n})$ in $\mathcal{H}_1$ follow the identical distribution modulo $\mathbf{T}\mathbb{Z}^d$ since $f_{i,n} \stackrel{\$}{\leftarrow} \mathbb{Z}_p$. Thus, $\hat{f}_{i,n}$ in tr and $\mathcal{H}_1$ are also statistically close.

For $\mathbf{r}_n$, it follows $\mathcal{D}_{\mathbb{Z}^d, \sqrt{n+1} \cdot \tau_1}^{\mu+\nu}$ in the real transcript tr , while it follows $\sum_{j=0}^n \mathcal{D}_{\mathbb{Z}^d, \tau_1}^{\mu+\nu}$ in the hybrid distribution $\mathcal{H}_1$. We note that these two distributions are statistically close due to Lemma 6.

Hybrid 2: We define the second hybrid distribution as follows.

$\mathcal{H}_2(\text{pp}, f, x, y) \rightarrow (\hat{\mathbf{u}}, \hat{\mathbf{t}}, \mathbf{z}, \mathbf{c}, \bar{\mathbf{f}}, \bar{\mathbf{r}})$

1. $(_, \text{op}) \leftarrow \text{Com}(\text{pp}, f)$, where $\text{op} = (\hat{\mathbf{F}}, \mathbf{R}, \hat{\mathbf{t}}, 1, 1)$.
2. $\mathbf{c} = (c_0, \dots, c_{n-1}) \stackrel{\$}{\leftarrow} \text{Ecd}(\mathcal{C})^n$.
3. $\mathbf{t}_j \stackrel{\$}{\leftarrow} R_q^\mu$ and redefine $\hat{\mathbf{t}}_j := \lfloor \mathbf{t}_j / q_h \rfloor$ for $0 \leq j < n$.
4. $\hat{f}_{0,n}^{(i,j)} \stackrel{\$}{\leftarrow} \text{REcd}_{\sigma_2}(\text{Dcd}(\hat{x}_i c_j \cdot \hat{f}_{i,j}))$
for $1 \leq i \leq m$, $0 \leq j < n$, and $\hat{\mathbf{x}} = (\hat{x}_0, \dots, \hat{x}_m)$.
5. $\hat{f}_{0,n}^* \stackrel{\$}{\leftarrow} \text{REcd}_{\sigma_2}(f_{0,n})$ for $f_{0,n} \stackrel{\$}{\leftarrow} \mathbb{Z}_p$.
6. $\hat{f}_{0,n} := \hat{f}_{0,n}^* + \sum_{i=1}^m \sum_{j=0}^{n-1} \hat{f}_{0,n}^{(i,j)}$.
7. $\hat{f}_{i,n}^{(j)} \stackrel{\$}{\leftarrow} \text{REcd}_{\sigma_3}(-\text{Dcd}(c_j \cdot \hat{f}_{i,j}))$ for $1 \leq i \leq m$ and $0 \leq j < n$.
8. $\hat{f}_{i,n}^* \stackrel{\$}{\leftarrow} \text{REcd}_{\sigma_3}(f_{i,n})$ for $f_{i,n} \stackrel{\$}{\leftarrow} \mathbb{Z}_p$ and $1 \leq i \leq m$.
9. $\hat{f}_{i,n} := \hat{f}_{i,n}^* + \sum_{j=0}^{n-1} \hat{f}_{i,n}^{(j)}$ for $1 \leq i \leq m$.
10. Redefine $\hat{\mathbf{f}}_n := (\hat{f}_{0,n}, \dots, \hat{f}_{m,n})$.
11. $\mathbf{r}_n^{(0)}, \dots, \mathbf{r}_n^{(n-1)}, \mathbf{r}_n^* \stackrel{\$}{\leftarrow} \mathcal{D}_{\mathbb{Z}^d, \tau_1}^{\mu+\nu}$.
12. Redefine $\mathbf{r}_n := \mathbf{r}_n^* + \sum_{j=0}^{n-1} \mathbf{r}_n^{(j)}$.
13. Redefine $\hat{\mathbf{t}}_n := \lfloor (\mathbf{A}\hat{\mathbf{f}}_n + \mathbf{B}\mathbf{r}_n) / q_h \rfloor$.
14. $\mathbf{z}^\top := \hat{\mathbf{x}}^\top \hat{\mathbf{F}}$, $\bar{\mathbf{f}} := \hat{\mathbf{F}} \begin{bmatrix} \mathbf{c} \\ 1 \end{bmatrix}$, $\mathbf{r} := \mathbf{R} \begin{bmatrix} \mathbf{c} \\ 1 \end{bmatrix}$, and $\hat{\mathbf{u}} := \lfloor \mathbf{D}\hat{\mathbf{t}} / q_s \rfloor$.

Fig. 7: Hybrid Distribution 2

The difference between $\mathcal{H}_1$ and $\mathcal{H}_2$ lies in how $\hat{\mathbf{t}}$ is generated, as described in the underlined parts in Fig. 7. We first note that the following equation holds

for both $\mathcal{H}_1$ and $\mathcal{H}_2$ by definition.

$$\bar{\mathbf{r}} = \sum_{j=0}^{n-1} c_j \cdot \mathbf{r}_j + \mathbf{r}_n = \sum_{j=0}^{n-1} (c_j \cdot \mathbf{r}_j + \mathbf{r}_n^{(j)}) + \mathbf{r}_n^{(0)}$$

Next, we note that the distribution of $(\mathbf{B}\mathbf{r}_j, c_j \cdot \mathbf{r}_j + \mathbf{r}_n^{(j)})$ and $(\mathbf{t}_j, c_j \cdot \mathbf{r}_j + \mathbf{r}_n^{(j)})$ are computationally indistinguishable if $\text{MLWE}_{q,\nu,\tau}$ is computationally hard, due to Theorem 1. This implies that the distribution of $(\hat{\mathbf{t}}, \bar{\mathbf{r}})$ in $\mathcal{H}_1$ and $\mathcal{H}_2$ are computationally indistinguishable, since $\mathbf{t}_j \leftarrow R_q^\mu$.

Hybrid 3: We define the third hybrid distribution as follows.

$\mathcal{H}_3(\text{pp}, x, y) \rightarrow (\hat{\mathbf{u}}, \hat{\mathbf{t}}, \mathbf{z}, \mathbf{c}, \bar{\mathbf{f}}, \bar{\mathbf{r}})$

1. $\mathbf{c} = (c_0, \dots, c_{n-1}) \xleftarrow{\$} \text{Ecd}(\mathcal{C})^n$.
2. $\mathbf{t}_j \xleftarrow{\$} R_q^\mu$ and define $\hat{\mathbf{t}}_j := \lfloor \mathbf{t}_j / q_h \rfloor$ for $0 \leq j < n$.
3. $f_{0,1}, \dots, f_{0,n-1} \xleftarrow{\$} \mathbb{Z}_p$ and $f_{0,0} := y - \sum_{j=1}^{n-1} f_{0,j} x^j$
4. $\hat{f}_{0,j}^{(0)} \xleftarrow{\$} \text{REcd}_{\sigma_0}(f_{0,j})$ for $0 \leq j < n$.
5. $\hat{f}_{0,j}^{(i)} \xleftarrow{\$} \text{REcd}_{\sigma_0}(0)$ for $1 \leq i \leq m$ and $0 \leq j < n$.
6. $\hat{f}_{0,j} := \hat{f}_{0,j}^{(0)} + \sum_{i=1}^m \hat{f}_{0,j}^{(i)}$ for $0 \leq j < n$.
7. $\hat{f}_{i,j} \xleftarrow{\$} \text{REcd}_{\sigma_1}(0)$ for $1 \leq i \leq m$ and $0 \leq j < n$.
8. $\hat{f}_{0,n}^{(i,j)} \xleftarrow{\$} \text{REcd}_{\sigma_2}(\text{Dcd}(\hat{x}_i c_j \cdot \hat{f}_{i,j}))$
for $1 \leq i \leq m$, $0 \leq j < n$, and $\hat{\mathbf{x}} = (\hat{x}_0, \dots, \hat{x}_m)$.
9. $\hat{f}_{0,n}^* \xleftarrow{\$} \text{REcd}_{\sigma_2}(f_{0,n})$ for $f_{0,n} \xleftarrow{\$} \mathbb{Z}_p$.
10. $\hat{f}_{0,n} := \hat{f}_{0,n}^* + \sum_{i=1}^m \sum_{j=0}^{n-1} \hat{f}_{0,n}^{(i,j)}$.
11. $\hat{f}_{i,n}^{(j)} \xleftarrow{\$} \text{REcd}_{\sigma_3}(-\text{Dcd}(c_j \cdot \hat{f}_{i,j}))$ for $1 \leq i \leq m$ and $0 \leq j < n$.
12. $\hat{f}_{i,n}^* \xleftarrow{\$} \text{REcd}_{\sigma_3}(f_{i,n})$ for $f_{i,n} \xleftarrow{\$} \mathbb{Z}_p$ and $1 \leq i \leq m$.
13. $\hat{f}_{i,n} := \hat{f}_{i,n}^* + \sum_{j=0}^{n-1} \hat{f}_{i,n}^{(j)}$ for $1 \leq i \leq m$.
14. Redefine $\hat{\mathbf{f}}_n := (\hat{f}_{0,n}, \dots, \hat{f}_{m,n})$
15. $\mathbf{r}_n^{(0)}, \dots, \mathbf{r}_n^{(n-1)}, \mathbf{r}_n^* \xleftarrow{\$} \mathcal{D}_{\mathbb{Z}^d, \tau_1}^{\mu+\nu}$.
16. Redefine $\mathbf{r}_n := \mathbf{r}_n^* + \sum_{j=0}^{n-1} \mathbf{r}_n^{(j)}$.
17. Redefine $\hat{\mathbf{t}}_n := \lfloor (\mathbf{A}\hat{\mathbf{f}}_n + \mathbf{B}\mathbf{r}_n) / q_h \rfloor$.
18. $\mathbf{z}^\top := \hat{\mathbf{x}}^\top \hat{\mathbf{F}}$, $\bar{\mathbf{f}} := \hat{\mathbf{F}} \begin{bmatrix} \mathbf{c} \\ 1 \end{bmatrix}$, $\mathbf{r} := \mathbf{R} \begin{bmatrix} \mathbf{c} \\ 1 \end{bmatrix}$, and $\hat{\mathbf{u}} := \lfloor \mathbf{D}\hat{\mathbf{t}} / q_s \rfloor$.

Fig. 8: Hybrid distribution 3

The difference between $\mathcal{H}_2$ and $\mathcal{H}_3$ lies in how $\hat{\mathbf{f}}_j = (\hat{f}_{0,j}, \dots, \hat{f}_{m,j})$ are sampled for $0 \leq j < n$. Since the distribution of $\hat{\mathbf{f}}_j$ only affects the distribution of $\bar{\mathbf{f}}$ and $\mathbf{z}$ in the output, it suffices to show that these two distributions in $\mathcal{H}_2$ and $\mathcal{H}_3$ are statistically close.

We first note that the distribution of $\hat{f}_{0,j}$ in $\mathcal{H}_2$ is statistically close to the distribution of $\hat{f}_{0,j}^{(0)} + \sum_{i=1}^m \hat{f}_{0,j}^{(i)}$ for $0 \leq j < n$, due to Lemma 6, which are sampled as follows.

$$\begin{aligned} \hat{f}_{0,j}^{(i)} &\stackrel{\$}{\leftarrow} \text{REcd}_{\sigma_1}(-\text{Dcd}(\hat{x}_i \cdot f_{i,j})) \text{ for } 1 \leq i \leq m \\ \hat{f}_{0,j}^{(0)} &\stackrel{\$}{\leftarrow} \text{REcd}_{\sigma_1}\left(f_{0,j} + \sum_{i=1}^m \text{Dcd}(\hat{x}_i \cdot f_{i,j})\right) \end{aligned}$$

Then, each component of $\bar{\mathbf{f}} = (\bar{f}_0, \dots, \bar{f}_m)$ and $\mathbf{z} = (z_0, \dots, z_n)$ can be written as follows in both $\mathcal{H}_2$ and $\mathcal{H}_3$.

$$\begin{aligned} \bar{f}_0 &= (\hat{f}_{0,n}^* + \sum_{j=0}^{n-1} c_j \cdot \hat{f}_{0,j}^{(0)}) + \sum_{i=1}^m \sum_{j=0}^{n-1} (\hat{f}_{0,n}^{(i,j)} + c_j \cdot \hat{f}_{0,j}^{(i)}) \\ \bar{f}_i &= \hat{f}_{i,n}^* + \sum_{j=0}^{n-1} (\hat{f}_{i,n}^{(j)} + c_j \cdot \hat{f}_{i,j}) \text{ for } 1 \leq i \leq m \\ z_j &= \hat{f}_{0,j}^{(0)} + \sum_{i=1}^m (\hat{f}_{0,j}^{(i)} + \hat{x}_i \cdot \hat{f}_{i,j}) \text{ for } 0 \leq j < n \\ z_n &= \hat{\mathbf{x}}^\top \bar{\mathbf{f}} - \sum_{j=0}^{n-1} c_j \cdot z_j \end{aligned}$$

We note that the distributions of $\sum_{i=1}^m (\hat{f}_{0,j}^{(i)} + \hat{x}_i \cdot \hat{f}_{i,j})$, $\sum_{j=0}^{n-1} (\hat{f}_{i,n}^{(j)} + c_j \cdot \hat{f}_{i,j})$, and $\sum_{i=1}^m \sum_{j=0}^{n-1} (\hat{f}_{0,n}^{(i,j)} + c_j \cdot \hat{f}_{0,j}^{(i)})$ in $\mathcal{H}_2$ and $\mathcal{H}_3$ are statistically close due to Lemma 9. Also, the distribution of $\hat{f}_{i,n}^*$ and $\hat{f}_{0,n}^*$ in $\mathcal{H}_2$ and $\mathcal{H}_3$ is identical by definition for $1 \leq i \leq m$. Finally, the distribution of $\hat{f}_{0,j}^{(0)}$ in $\mathcal{H}_2$ and $\mathcal{H}_3$ is also identical for $0 \leq j < n$ since $(\text{Dcd}(\hat{f}_{0,0}^{(0)}), \dots, \text{Dcd}(\hat{f}_{0,n-1}^{(0)}))$ follows the uniform distribution over the set in $\mathbb{Z}_p^n$ where the inner product with $(1, x, \dots, x^{n-1})$ equals y . As a result, $\mathcal{H}_2$ and $\mathcal{H}_3$ are statistically close. We remark that $\mathcal{H}_3$ does not take f as input, so $\mathcal{H}_3$ becomes a simulator for the hiding property. $\square$

C.7 Proof of Theorem 8

Proof. We analyze the computational complexity of the prover and verifier, and the communication cost of Π_{Eval} in Fig. 3.

Prover. For the prover's complexity, we measure the computation cost of **Com** and **Eval.P** together. The time complexity of **Com** is dominated by the computation of $\mathbf{t}_j = \mathbf{A}\hat{\mathbf{f}}_j + \mathbf{B}\mathbf{r}_j$ for $0 \leq j < n$ in Fig. 2. More precisely, the computation of $\mathbf{A}\hat{\mathbf{f}}_j$ for $0 \leq j < n$ determines the overall complexity, which is $O(\mu mn) = O(\mu N)$, since $\mathbf{A} \in R_q^{\mu \times (m+1)}$ and $\hat{\mathbf{f}}_j \in R^{m+1}$. The parameter μ determines the computational hardness of $\text{MSIS}_{\mu, 2B_\alpha B_c B_j}$ and is chosen according to the heuristic in [MR09, GN08], which gives $O(\mu) = O(\log^2 N)$. Therefore, the time complexity of the commitment algorithm is $O(N \log^2 N)$. The time complexity

of **Eval.P** is dominated by the computation of $\bar{\mathbf{f}}$ in Fig. 3, which takes $O(mn) = O(N)$ complexity, as $\hat{\mathbf{F}} \in R^{(m+1) \times (n+1)}$.

Verifier. The time complexity of **Eval.V** is dominated by the computation of $\bar{\mathbf{h}}$ and the computation of $\mathbf{D}\hat{\mathbf{t}}$ in Fig. 3. The former takes $O(\mu m)$ complexity, and the latter takes $O(\kappa \mu n)$ complexity, since $\mathbf{A} \in R_q^{\mu \times (m+1)}$ and $\mathbf{D} \in R_q^{\kappa \times \mu(n+1)}$. Since both μ and κ are selected following the heuristic in [MR09, GN08], it holds that $O(\mu) = O(\kappa) = O(\log^2 N)$. Thus, the time complexity of the verifier is $O(m \log^2 N + n \log^4 N)$.

Communication. The communication complexity is dominated by the size of $\bar{\mathbf{f}} \in R^{m+1}$ and $\hat{\mathbf{t}} \in R^{\mu n}$ in Fig. 3. Thus, the total communication complexity is $O(m + n \log^2 N)$, assuming $O(\mu) = O(\log^2 N)$.

Optimum. The complexity of verification and communication depends on the choice of m and n , which satisfy $mn = N$. Both complexities achieve their optimal values when $m = O(\sqrt{N} \log N)$ and $n = O(\sqrt{N} / \log N)$, resulting in a verification complexity of $O(\sqrt{N} \log^3 N)$ and a communication complexity of $O(\sqrt{N} \log N)$. □

C.8 Proof of Theorem 9

Proof. Relaxed completeness and public reducibility directly follow from observation. For knowledge soundness, it suffices to prove that coordinate-wise special soundness holds, due to Lemma 10. Below, we prove the coordinate-wise special soundness of (**Batch.P**, **Batch.V**).

Let $(\mathbf{x}^{(k)}, \mathbf{w}^{(k)}) \in \mathcal{R}_{\text{PCS}}^{(\underline{B}'_j, \underline{B}'_o, \underline{B}'_c)}$ be the reduced instance-witness pairs generated from a challenge $\alpha^{(k)}$ for $0 \leq k \leq \ell$, where $\mathbf{x}^{(k)} = (\hat{\mathbf{u}}^{(k)}, x, y^{(k)})$, $\mathbf{w}^{(k)} = (f^{(k)}, \hat{\mathbf{F}}^{(k)}, \mathbf{R}^{(k)}, \hat{\mathbf{t}}^{(k)}, \mathbf{c}^{(k)}, 1)$, and $\alpha^{(k)}$ differs from $\alpha^{(\ell)}$ only in the k -th coordinate. Also, define $\mathbf{s}^{(k)}$ and $\mathbf{h}_j^{(k)}$ as follows for $0 \leq k \leq \ell$ and $0 \leq j < n$.

$$\begin{aligned} - \mathbf{s}^{(k)} &:= \mathbf{D}\hat{\mathbf{t}}^{(k)} - q_s \hat{\mathbf{u}}^{(k)} \pmod{q_o} \\ - \mathbf{h}_j^{(k)} &:= \mathbf{A}\hat{\mathbf{f}}_j^{(k)} + \mathbf{B}\mathbf{r}_j^{(k)} - q_h c_j^{(k)} \hat{\mathbf{t}}_j^{(k)} \pmod{q} \end{aligned}$$

Next, define $\mathbf{w}_k = (f_k, \hat{\mathbf{F}}_k, \mathbf{R}_k, \hat{\mathbf{t}}_k, \mathbf{c}_k, \alpha_k)$ as follows for $0 \leq k < \ell$ and $0 \leq j < n$.

$$\begin{aligned} - \alpha_k &:= \alpha_k^{(k)} - \alpha_k^{(\ell)} \\ - \mathbf{c}_k &:= \mathbf{c}_k^{(k)} \odot \mathbf{c}_k^{(\ell)} \\ - f_k &:= \text{Dcd}(\alpha_k)^{-1} \cdot (f^{(k)} - f^{(\ell)}) \\ - \hat{\mathbf{t}}_k &:= \hat{\mathbf{t}}_k^{(k)} - \hat{\mathbf{t}}_k^{(\ell)} \\ - \hat{\mathbf{f}}_{k,j} &:= c_j^{(\ell)} \cdot \hat{\mathbf{f}}_j^{(k)} - c_j^{(k)} \cdot \hat{\mathbf{f}}_j^{(\ell)} \\ - \mathbf{r}_{k,j} &:= c_j^{(\ell)} \cdot \mathbf{r}_j^{(k)} - c_j^{(k)} \cdot \mathbf{r}_j^{(\ell)} \end{aligned}$$

Also, define $\mathbf{s}_k$ and $\mathbf{h}_{k,j}$ as follows for $0 \leq k \leq \ell$ and $0 \leq j < n$.

$$\begin{aligned}
- \mathbf{s}_k &:= \mathbf{s}^{(k)} - \mathbf{s}^{(\ell)} \\
- \mathbf{h}_{k,j} &:= c_j^{(\ell)} \cdot \mathbf{h}_j^{(k)} - c_j^{(k)} \cdot \mathbf{h}_j^{(\ell)}
\end{aligned}$$

Then, the following holds for $0 \leq k < \ell$ and $0 \leq j < n$.

$$\begin{aligned}
- \mathbf{s}_k &= \mathbf{D}\hat{\mathbf{t}}_k - q_s \alpha_k \cdot \hat{\mathbf{u}}_k \pmod{q_o} \\
- \mathbf{h}_{k,j} &= \mathbf{A}\hat{\mathbf{f}}_{k,j} + \mathbf{B}\mathbf{r}_{k,j} - q_h c_{k,j} \hat{\mathbf{t}}_{k,j} \pmod{q} \\
- \|\mathbf{s}_k\|_2 + \|\hat{\mathbf{t}}_k\|_2 &\leq 2\hat{B}'_o = \hat{B}'_o \\
- \|c_{k,j}\|_1 &= \|c_j^{(k)} \cdot c_j^{(\ell)}\|_1 \leq (\hat{B}'_c)^2 = \hat{B}'_c \\
- \|\alpha_k\|_1 &= \|\alpha_k^{(k)} - \alpha_k^{(\ell)}\|_1 \leq 2B_\alpha = \hat{B}'_\alpha \\
- \|\hat{\mathbf{f}}_{k,j}\|_2 + \|\mathbf{r}_{k,j}\|_2 + \|\mathbf{h}_{k,j}\|_2 &\leq 2\hat{B}'_c \hat{B}' = \hat{B}'
\end{aligned}$$

Also, the following holds for $0 \leq k < \ell$, $0 \leq i < m$, and $0 \leq j < n$.

$$\begin{aligned}
f_{k,i,j} &= \text{Dcd}(c_{k,j}\alpha_k)^{-1} \cdot \text{Dcd}(\hat{f}_{k,i,j}) \\
&= \text{Dcd}(c_{k,j}\alpha_k)^{-1} \cdot \text{Dcd}(c_j^{(\ell)} \hat{f}_{i,j}^{(k)} - c_j^{(k)} \hat{f}_{i,j}^{(\ell)}) \\
&= \text{Dcd}(\alpha^{(k)})^{-1} \cdot (f_{i,j}^{(k)} - f_{i,j}^{(\ell)})
\end{aligned}$$

Therefore, $(\mathbf{x}_k, \mathbf{w}_k) \in \mathbf{R}_{\text{PCS}}^{(B'_j, B'_o, B'_c, B'_\alpha)}$ for $0 \leq k < \ell$. $\square$

C.9 Proof of Theorem 10

Proof. We analyze the computational complexity of the prover and verifier, and the communication cost of $\Pi_{\text{BatchEval}}$ in Fig. 3.

Prover. The overall prover complexity follows $O(t\mu N)$, where $O(\mu) = O(\log^2(tN))$, following the heuristic in [MR09, GN08]. Thus, the total time complexity is $O(\ell N \log^2(\ell N))$.

Verifier. The verifier complexity of Π_{Batch} is dominated by the computation of $\mathbf{u}$, which takes $O(\ell\kappa)$ complexity. The verification of Π_{Eval} takes $O(\mu m + \kappa\mu n)$, as shown in the analysis in Theorem 8. Thus, the overall verification complexity is $O(m \log^2(\ell N) + n \log^4(\ell N) + t \log^2(\ell N))$, assuming $O(\mu) = O(\kappa) = O(\log^2(\ell N))$.

Communication. The communication complexity of Π_{Batch} is dominated by the total size of commitments, which scales as $O(\kappa t)$. The communication complexity of Π_{Eval} is $O(m + \mu n)$, as shown in the analysis in Theorem 8. Therefore, the overall communication complexity is $O(m + n \log^2(\ell N) + t \log^2(\ell N))$, assuming $O(\mu) = O(\kappa) = O(\log^2(\ell N))$.

Optimum. The complexity of verification and communication depends on the choice of m and n , which satisfy $mn = N$. Both complexities achieve their optimal values when $m = O(\sqrt{N} \log(\ell N))$ and $n = O(\sqrt{N} / \log(\ell N))$, resulting in a verification complexity of $O(\sqrt{N} \log^3(\ell N) + t \log^2(\ell N))$ and communication complexity of $O(\sqrt{N} \log(\ell N) + \ell \log^2(\ell N))$. $\square$